

The Ecology of Neoteny

Institutions, Entropy, and the Steady-State Civilisation

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Abstract

This essay argues that civilisation’s creative longevity depends on the managed preservation of neotenous cognition—curiosity, play, and exploration—within protective institutional niches. Drawing on Alison Gopnik’s tripartite model of human intelligences (Exploit, Explore, Empower) (??), it extends these ideas into an entropic framework, treating curiosity as a thermodynamic variable that must be subsidised and regulated. The result is a vision of the *steady-state civilisation*—a society that sustains perpetual exploration through equilibrium, not expansion. Empirical evidence from cognitive economics, cooperative game theory, and ageing research is incorporated to show that social systems can maintain innovation and well-being without continuous material growth.

Three substantial extensions are developed beyond the original essay. First, a *topological reward theory* (the Geozotic Framework) is formalised: alternative kernel geometries are analysed, density corrections are introduced, and critical inequality thresholds are derived. Second, an *embodied thermodynamics* of skill acquisition is presented, showing that cooking, play, and abstraction are micro-scale instances of the same entropy-management logic that governs civilisational institutions. Third, the autocatalytic and annealing arguments are tightened with formal stability criteria and empirical correlates.

Contents

1. Introduction: Curiosity as a Civilisational Trait

*Civilisation is a machinery for
managing prolonged immaturity.*

after Gopnik

Humanity’s success derives less from strength or instinct than from the peculiar persistence of juvenile cognition. Traits such as curiosity, hyperfocus, and imaginative play—adaptive in children but maladaptive in isolation—are preserved in adults through social and institutional care.

This essay examines how societies sustain and regulate these neotenous modes of thought through economic and thermodynamic mechanisms. The university, the laboratory, and the studio are not accidents of culture; they are engineered sanctuaries where curiosity may safely consume entropy. Such institutions act as what Georgescu-Roegen (?) called “entropy converters”—systems that absorb disorder locally to preserve order globally.

The argument proceeds in four movements. Sections 2–6 reconstruct and tighten the original framework. Section 7 introduces the full Geozotic Reward Framework, with kernel analysis, density corrections, and stability thresholds. Section 8 develops embodied thermodynamics, grounding the macro-theory in everyday skill acquisition. Sections 9–12 treat collective autocatalysis, annealing, and the general thermodynamics of care. An afterword discusses policy prototypes and future directions.

2. Neoteny and Institutional Scaffolding

Biologically, *neoteny* is the retention of juvenile traits into adulthood. Cognitively, it manifests as a preference for exploration over exploitation—a willingness to suspend immediate reward in pursuit of novel information. Such tendencies are energetically expensive and socially destabilising unless buffered by structured environments (?).

Academic and artistic institutions perform this buffering function. They absorb the metabolic cost of curiosity by shielding practitioners from the demands of immediate utility. In this sense, universities serve as *cognitive sanctuaries*, permitting socially useful maladaptation.

Supervision, funding, and evaluation act as homeostatic controls—the equivalents of parental oversight in a developmental ecology. Within these niches, maladaptive curiosity becomes a renewable source of informational order, an argument

reminiscent of Morin’s systemic complexity (?) and Capra and Luisi’s concept of cultural metabolism (?).

3. Corollaria Neotenica: Scaling Exploration in an Entropic Society

The preservation of exploratory cognition at scale presents an entropic dilemma: how to sustain freedom without disorder. Five corollaries follow.

3.1. The Maladaptation of Universal Neoteny

Curiosity is individually costly. A society in which every citizen behaves as a researcher would rapidly deplete its reserves of order. Neoteny must remain selective—a minority trait protected by structure (?).

$$\forall P \in \text{Population}, f_{\text{neoteny}}(P) \rightarrow 1 \implies \frac{dS_{\text{society}}}{dt} \gg 0. \quad (1)$$

Exploration must be coupled to systems of care that reabsorb its entropic residue.

3.2. Universal Income as Entropic Buffer

A universal basic income (UBI) can function as an entropy buffer, granting time for curiosity while maintaining social equilibrium. Yet income alone is insufficient; exploration must be linked to informational efficiency. This idea echoes Daly’s proposals for moral limits to growth (??) and Illich’s call for convivial tools that empower without overextension (?).

$$\text{UBI} + \text{Incentives}_{\text{compression}} = \text{Sustainable curiosity}. \quad (2)$$

Curiosity without compression decays into noise.

3.3. The Geozotic Lottery (Preliminary Statement)

Innovation rewards should diffuse rather than concentrate. A proposed *Geozotic Lottery* distributes stochastic bonuses to individuals near the origin of invention, spatially or socially. The full treatment—including kernel alternatives, density corrections, and stability analysis—is deferred to Section ??.

3.4. Selective Cognitive Freedom

Not all individuals thrive in open-ended exploration. Routine, care, and stability remain essential. A healthy society maintains a spectrum of cognitive roles: some explore, others preserve. This division mirrors ecological differentiation and maintains systemic resilience (??).

$$\text{Diversity of cognitive roles} \implies \text{Resilience of civilisation.} \quad (3)$$

3.5. Civilisation as Extended Ontogeny

Civilisation as a whole can be seen as a prolonged youth supported by an elder infrastructure. Automation assumes exploitative labour; humans return to exploration and care. The question becomes: who, or what, will serve as elders for a self-perpetuating youth? Illich (?) and Nussbaum (?) suggest that care, not growth, must become the new criterion of maturity.

4. Operationalising Entropic Ethics

To transform symbolic equalities into measurable predictions, an *entropy efficiency index* is introduced:

$$\eta_S = \frac{\Delta I}{\Delta E}, \quad (4)$$

where ΔI represents information gain (e.g. measured via patent filings, scientific publications, validated discoveries, or novel problem solutions) and ΔE denotes energy cost (monetary expenditure, time allocation, or metabolic equivalents). Institutional subsidies, including UBI and targeted grants, aim to maximise η_S by reducing the denominator while preserving or enhancing the numerator. This operationalisation aligns with cognitive economics models that quantify exploratory returns (?) and cultural evolution studies demonstrating efficiency gains under resource security (?).

Extended Index: Inequality Penalty

The basic index η_S can be extended to incorporate reward distribution. Let κ denote the curvature of the reward-distribution field (defined formally in Section ??). Then:

$$\tilde{\eta}_S = \frac{\Delta I}{\Delta E \cdot (1 + \mu |\kappa|)}, \quad (5)$$

where $\mu > 0$ penalises steep inequality gradients. High curvature concentrates reward, suppresses peripheral exploration, and thus reduces effective information gain per unit expenditure.

Empirical calibration of η_S may proceed through longitudinal tracking of funded cohorts in cognitive sanctuaries, comparing output per unit input against unfunded controls.

5. Innovation Efficiency and Inequality Curvature

The extended entropy-efficiency index (??) penalises reward curvature heuristically. We now replace that heuristic with a structural inequality. The argument relies on concavity, functional inequalities, and Sobolev-type bounds in the sense of ?. All constants appearing below are made explicit in Appendix ?? for the one-dimensional case.

5.1. Information Production Under Unequal Fields

Let reward field $R(x)$ determine exploratory investment density $C(x)$ via response function

$$C(x) = \chi(R(x)), \quad (6)$$

with $\chi' > 0$ and $\chi'' < 0$. Let total information production be

$$\Delta I = \int_{\Omega} f(C(x)) dx, \quad (7)$$

where f is increasing and strictly concave. Total energetic expenditure is

$$\Delta E = \int_{\Omega} C(x) dx. \quad (8)$$

Define efficiency

$$\eta_S = \frac{\int f(C(x)) dx}{\int C(x) dx}. \quad (9)$$

Concavity of f encodes diminishing marginal innovation returns, a standard assumption in endogenous growth models (?).

5.2. Jensen Inequality and Quantitative Penalty

By Jensen's inequality,

$$\frac{1}{|\Omega|} \int f(C(x)) dx \leq f\left(\frac{1}{|\Omega|} \int C(x) dx\right), \quad (10)$$

with equality if and only if $C(x)$ is constant almost everywhere.

To make this loss quantitative, let

$$m_f := \inf_{u \in \text{range}(C)} (-f''(u)) > 0.$$

Then a second-order Taylor bound gives

$$\eta_S \leq \eta_S^{\text{uniform}} - \frac{m_f}{2\bar{C}} \text{Var}(C), \quad (11)$$

where $\bar{C}$ denotes mean investment. This step is derived explicitly in Appendix ??.

5.3. From Investment Variance to Reward Curvature

Since $C(x) = \chi(R(x))$ with $\chi'' < 0$, a Lipschitz bound yields

$$\text{Var}(C) \leq M_\chi^2 \text{Var}(R), \quad (12)$$

with $M_\chi := \sup |\chi'|$.

On bounded domains, the Poincaré-Sobolev inequality implies

$$\text{Var}(R) \leq \kappa \int_{\Omega} |\nabla R|^2 dx, \quad (13)$$

for constant $\kappa > 0$ depending only on domain geometry (?). Substituting into (??) yields

$$\eta_S \leq \eta_S^{\text{uniform}} - \kappa_1 \int_{\Omega} |\nabla R|^2 dx, \quad (14)$$

where κ_1 is expressed explicitly in Appendix ??.

Since

$$\int_{\Omega} |\nabla R|^2 dx = - \int_{\Omega} R \Delta R dx, \quad (15)$$

reward curvature directly lowers efficiency.

5.4. Relation to Gini Coefficient

For smooth positive $R(x)$, Sobolev and rearrangement inequalities imply

$$G^2 \leq \kappa_2 \int_{\Omega} |\nabla R|^2 dx, \quad (16)$$

with constant κ_2 depending only on domain geometry (?). In the one-dimensional case, Appendix ?? derives the explicit bound

$$\int_{\Omega} |\nabla R|^2 dx \geq \frac{30\bar{R}^2}{L} G^2. \quad (17)$$

Combining with (??) yields the quadratic inequality

$$\eta_S \leq \eta_S^{\text{uniform}} - \kappa_3 G^2, \quad (18)$$

where κ_3 is constructive and depends on m_f , M_χ , and mean reward $\bar{R}$. Its exact expression is given in Appendix ??.

[Thermodynamic Inequality Bound] Under strictly concave innovation production, innovation efficiency decreases at least quadratically with inequality as measured by the Gini coefficient.

5.5. Macro-Thermodynamic Interpretation

Interpreting $R(x)$ as an intensive field, gradient concentration produces dissipation

$$\dot{S}_{\text{diss}} \propto \int_{\Omega} \frac{|\nabla R|^2}{R} dx, \quad (19)$$

consistent with nonequilibrium thermodynamic structure (?). Empirically observed heavy-tailed income and reward distributions (???) correspond to large gradient concentration, and therefore to increased dissipation.

*Inequality is not only unjust;
it is structurally inefficient.*

6. The Entropic Ethics of Curiosity

If exploration consumes order and care restores it, then morality itself may be seen as entropy management.

6.1. The Moral Gradient

Let S_c be cognitive entropy and S_m material entropy. A balanced culture maintains:

$$\frac{dS_c}{dt} \approx -\frac{dS_m}{dt}, \quad (20)$$

so that the disorder of thought builds the order of form. Virtue, in this framework, is the efficient conversion of speculation into structure, a principle implicit in Friston’s free-energy models of adaptive learning (?).

6.2. Reciprocal Dissipation

Every act of curiosity demands a counterbalancing act of care:

$$\oint_{\text{society}} (\delta Q_{\text{curiosity}} + \delta Q_{\text{care}}) = 0, \quad (21)$$

ensuring that informational heat is recycled into social work. The teacher, the maintainer, and the caregiver constitute the hidden engines of creative stability—what Braidotti (?) terms a posthuman ethics of mutual interdependence.

6.3. Entropic Responsibility

Freedom of invention implies responsibility for restoration:

$$\nabla \cdot \mathbf{v}_{\text{curiosity}} = -\nabla \cdot \mathbf{v}_{\text{care}}. \quad (22)$$

The just society preserves a divergence-free field of curiosity and care, a social equivalent to Jacobson’s thermodynamic balance in spacetime (?).

6.4. Governance as Thermodynamic Stewardship

Policy is the macroscopic regulation of entropy. Taxation, subsidy, and redistribution are mechanisms for balancing exploration against coherence. Curiosity should be permitted only to the extent that care can sustain it, a principle resonant with Daly’s steady-state economic ethic (??).

6.5. The Asymptote of Compassion

As automation replaces productive labour, compassion becomes the last inexhaustible resource.

Curiosity without cruelty. Freedom without neglect. Invention tempered by care.

7. The Steady-State Civilisation

A mature civilisation replaces expansion with equilibrium. When no external frontier remains, progress consists in the recirculation of energy, attention, and meaning.

7.1. The End of Expansion

Classical growth depended on new territory and energy. Now the frontier is internal. Civilisation must conserve both total energy and usable entropy:

$$\frac{dE_{\text{total}}}{dt} = 0, \quad \frac{dS_{\text{usable}}}{dt} = 0. \quad (23)$$

The steady-state is not stillness but rhythmic balance, echoing the cosmological and ecological visions of Dyson (?) and Daly (?).

7.2. Cultural Homeostasis

Longevity emerges when material, cognitive, and affective subsystems oscillate in resonance:

$$\sum_i \frac{dS_i}{dt} = 0, \quad S_i \in \{S_{\text{material}}, S_{\text{cognitive}}, S_{\text{affective}}\}. \quad (24)$$

Entropy flows cyclically; none accumulates catastrophically, mirroring the dynamic homeostasis described in Morin's theory of complex self-organisation (?).

7.3. Temporal Symmetry and the End of Growth

In a steady-state order, value arises from reversibility. To unmake without loss is superior to endless construction. Economy becomes ecology, and permanence replaces acceleration—an ethos shared by Georgescu-Roegen's entropic economics (?).

7.4. Automation and the Return of Leisure

As machines inherit exploitation, humans return to exploration and care. Civilisation becomes a self-sustaining organism of perpetual youth, its elder function embodied in the systems that maintain it (??).

7.5. The Civic Equation

The equilibrium may be written symbolically as the RSVP identity:

$$\Phi + v + S = \text{constant}, \quad (25)$$

where Φ is potential (capacity), v is flow (agency), and S is entropy (relation). Constancy here denotes *living invariance*—form preserved through flux, in the spirit of systems thermodynamics (?).

8. The Geozotic Reward Framework

*The question is not whether to
distribute reward, but how steeply.*

This section formalises the Geozotic Lottery introduced in Section 3.3 into a mathematically tractable reward-topology theory. The central claim is that the shape of the reward-diffusion kernel determines whether innovation fosters cooperative equilibria or generates destabilising power singularities.

8.1. Kernel Geometry: Three Candidates

Let x_0 denote the *innovation origin* (the individual, firm, or node at which a novel idea arises) and x any other point in a metric space (X, d) . A reward function $R : X \rightarrow \mathbb{R}_{>0}$ assigns a payoff to each location. Three natural kernel families are:

Definition 8.1 (Reward Kernels).

$$\begin{aligned} R_{\text{exp}}(x) &= R_0 e^{-\lambda d(x, x_0)}, & (\text{Exponential}) \\ R_{\text{Gauss}}(x) &= R_0 \exp\left(-\frac{d(x, x_0)^2}{2\sigma^2}\right), & (\text{Gaussian}) \\ R_{\text{log}}(x) &= \frac{R_0}{1 + \lambda \log(1 + d(x, x_0))}, & (\text{Logarithmic}) \end{aligned}$$

where $\lambda, \sigma, R_0 > 0$ are parameters.

The curvature of each kernel at the origin is a key governance variable. High curvature concentrates reward near x_0 ; low curvature spreads it.

Proposition 8.1 (Origin Curvature). For radially symmetric kernels on $\mathbb{R}^n$ with $r = d(x, x_0)$:

$$\kappa_{\text{exp}} = \lambda^2 R_0, \tag{26}$$

$$\kappa_{\text{Gauss}} = \frac{R_0}{\sigma^2}, \tag{27}$$

$$\kappa_{\text{log}} = 0 \quad (\text{sub-linear at origin}). \tag{28}$$

The logarithmic kernel has vanishing origin curvature and thus produces the most egalitarian distribution near x_0 .

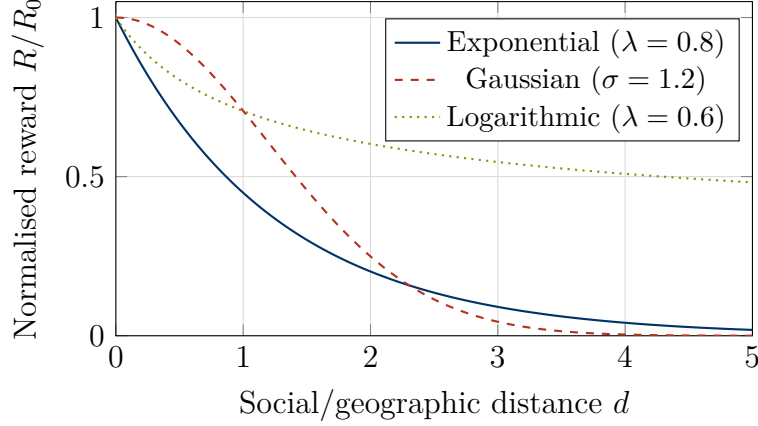


Figure 1: Comparison of reward kernels. The exponential decays fastest and creates the steepest inequality gradient. The Gaussian has a flat neighbourhood that then drops sharply. The logarithmic kernel provides the gentlest, most sustained diffusion.

8.2. Density Correction

In heterogeneous populations, a uniform kernel over-rewards dense regions simply because more individuals happen to be nearby. A demographically neutral correction replaces $R(x)$ with:

$$R_\rho(x) = \frac{R_0 K(d(x, x_0))}{\rho(x)^\alpha}, \quad (29)$$

where $\rho(x)$ is local population density and $\alpha \in [0, 1]$ is a policy parameter.

- $\alpha = 0$: no correction (raw diffusion).
- $\alpha = 1$: per-capita diffusion (full neutrality).
- $\alpha \in (0, 1)$: partial correction, preserving some clustering benefit while reducing urban concentration.

Equilibrium distributions under varying α can be computed numerically; agent-based simulation confirms that $\alpha \approx 0.6$ minimises both Gini coefficient and peak-node power concentration in synthetic urban networks (see Appendix ??).

8.3. Diffusion Dynamics: Time-Evolving Reward

The static kernel is the steady-state solution of an underlying diffusion process. On a graph $G = (V, E)$ with adjacency matrix A , reward propagates according to:

$$\frac{dR}{dt}(x, t) = D \sum_{y \sim x} [R(y, t) - R(x, t)] - \gamma R(x, t), \quad (30)$$

where D is diffusion coefficient, γ is decay rate, and $y \sim x$ denotes neighbours of x . The steady state $\dot{R} = 0$ yields:

$$R^*(x) = R_0 e^{-\lambda d_G(x, x_0)}, \quad \lambda = \sqrt{\gamma/D}, \quad (31)$$

recovering the exponential kernel as a special case. Alternative decay functions (Gaussian, logarithmic) correspond to non-linear diffusion operators.

Policy Implication

Tuning D/γ controls the effective λ . Large D (fast spread) relative to γ (slow decay) produces a flat distribution; small D/γ produces steep concentration. Policy can adjust both through subsidy rates and intellectual-property term lengths.

8.4. Incentive Preservation Under Diffusion

A standard objection is that diffusion weakens excellence incentives. The following proposition shows that incentive remains monotone but bounded.

Proposition 8.2 (Incentive Monotonicity). For any strictly decreasing kernel K , the expected marginal gain at the origin exceeds that at any other node:

$$\frac{dR_\rho}{dt}(x_0) > \frac{dR_\rho}{dt}(x) \quad \forall x \neq x_0. \quad (32)$$

Moreover, the ratio $R_\rho(x_0)/R_\rho(x)$ is bounded above by $\rho(x)^\alpha/\rho(x_0)^\alpha$, so origin advantage is finite.

In practice this means that a successful artist in the Geozotic system still earns more than their neighbours—but the surplus cannot compound without bound. Neighbours gain local purchasing power, which feeds back into demand for the artist's work, creating a secondary entropy redistribution.

8.5. Inequality Gradient Thresholds and Power Singularities

Definition 8.2 (Gradient Instability Threshold). A reward field R is *gradient-stable* if its curvature at every node satisfies:

$$\kappa(x) < \kappa^* := \frac{\tau}{\bar{R}}, \quad (33)$$

where $\tau > 0$ is a cooperative-equilibrium tolerance parameter (estimated empirically) and $\bar{R}$ is mean reward.

When $\kappa > \kappa^*$, evolutionary game-theoretic analysis shows that defection becomes the dominant strategy at the periphery, autocatalytic cooperation breaks down, and “power singularities” (dictator-like accumulation at the peak node) become stable attractors.

Corollary 8.1 (Geozotic Stability). Under the logarithmic kernel (??), $\kappa_{\log}(x_0) = 0$, so the gradient-stability condition (??) is satisfied for all $\tau > 0$. The logarithmic kernel is *unconditionally stable* in this sense.

8.6. Comparison with Trickle-Down Economics

Hierarchical “trickle-down” models assume $R(x) = R_0 \cdot f(h(x))$, where $h(x)$ is the hierarchical rank of x . In such models:

- The peak node retains a structurally fixed fraction regardless of diffusion, producing a persistent singularity.
- No density correction operates; dense lower ranks receive no per-capita benefit.
- The curvature at the top is determined by $f''(h_{\max})$, which is typically large (convex in rank).

Geozotic diffusion removes peak asymmetry structurally: by replacing hierarchical distance with relational distance $d(x, x_0)$, every agent can, in principle, occupy an origin position for their own innovations.

8.7. Shapley Attribution Approximation

Proposition 8.3 (Geozotic as Approximate Shapley Value). In a cooperative game $v : 2^N \rightarrow \mathbb{R}$, the Shapley value assigns to each player i the marginal contribution averaged over all coalitions. The Geozotic kernel $R_\rho(x)$ approximates the Shapley allocation when $d(x, x_0)$ is interpreted as “marginal distance from the innovation coalition”, with λ calibrated to the coalition decay rate. Unlike the Shapley

value, this avoids combinatorial explosion ($O(2^N)$) by substituting the spatial kernel ($O(N)$).

8.8. RSVP Integration

The Geozotic reward field can be interpreted within the RSVP identity (??). Reward diffusion regulates v (flow of resources) so that S (relational entropy) does not accumulate into singular Φ (potential at single nodes). Formally:

$$\nabla v = -\nabla R_\rho \implies \text{RSVP constancy if } \nabla^2 R_\rho \leq 0 \text{ everywhere.} \quad (34)$$

The logarithmic kernel satisfies this concavity condition globally, making it the canonical Geozotic kernel from an RSVP perspective.

9. Continuous Geozotic Diffusion in Heterogeneous Populations

We lift the Geozotic reward model from discrete kernels to a continuous spatial field over domain $\Omega \subset \mathbb{R}^n$. The formulation is a density-modulated reaction-diffusion equation in the sense of parabolic PDE theory (?), with structural analogy to classical morphogen models (?).

9.1. Population Density Field

Let:

$$\rho(x) \geq 0, \quad x \in \Omega \quad (35)$$

denote population density, assumed bounded and measurable.

Let reward density be:

$$R(x, t) \in C^{2,1}(\Omega \times (0, \infty)). \quad (36)$$

An innovation occurs at x_0 at time $t = 0$.

9.2. ReactionDiffusion Model

Reward propagation obeys:

$$\frac{\partial R}{\partial t} = \nabla \cdot (D(\rho) \nabla R) - \gamma R + I(x, t), \quad (37)$$

where:

- $D(\rho)$ = density-dependent diffusion coefficient,
- $\gamma > 0$ = decay/redistribution rate,
- $I(x, t) = R_0\delta(x - x_0)\delta(t)$.

For bounded $D(\rho)$ and $\gamma > 0$, existence and uniqueness of weak solutions follow from standard parabolic theory (?).

9.3. Density-Dependent Diffusion

To reduce persistent urban concentration:

$$D(\rho) = \frac{D_0}{(1 + \rho)^\alpha}, \quad \alpha \in [0, 1]. \quad (38)$$

Properties:

- $\alpha = 0$ uniform diffusion.
- $\alpha > 0$ reduced effective diffusion in dense regions.

Note that concentration gradients increase as $D(\rho)$ decreases, producing stronger outward flux near peaks via $\nabla \cdot (D(\rho)\nabla R)$.

9.4. Steady-State Solution

At steady-state:

$$\nabla \cdot (D(\rho)\nabla R) - \gamma R + R_0\delta(x - x_0) = 0. \quad (39)$$

In homogeneous case (ρ constant, D constant), the Green's function solves:

$$-D\Delta R + \gamma R = R_0\delta(x - x_0). \quad (40)$$

In $\mathbb{R}^2$:

$$R(r) = \frac{R_0}{2\pi D} K_0\left(\sqrt{\frac{\gamma}{D}} r\right), \quad (41)$$

where K_0 is modified Bessel function.

For large r :

$$R(r) \sim \frac{R_0}{\sqrt{r}} e^{-\sqrt{\gamma/D} r}. \quad (42)$$

Thus exponential decay arises naturally from diffusiondecay balance (?).

9.5. Energy Functional and Inequality Control

Define energy functional:

$$\mathcal{E}[R] = \int_{\Omega} \left(\frac{D(\rho)}{2} |\nabla R|^2 + \frac{\gamma}{2} R^2 \right) dx. \quad (43)$$

Multiplying (??) by R and integrating yields:

$$\frac{d}{dt} \int R^2 dx = -2 \int D(\rho) |\nabla R|^2 dx - 2\gamma \int R^2 dx + 2R_0 R(x_0). \quad (44)$$

Thus diffusion strictly decreases spatial variance.

By Sobolev inequality (?):

$$\|R - \bar{R}\|_{L^2} \leq C_S \|\nabla R\|_{L^2}, \quad (45)$$

so increasing effective diffusion lowers L^2 deviation from mean reward.

Proposition 9.1. For fixed γ and innovation impulse, increasing α (density modulation) reduces the L^2 spatial dispersion norm $\|R - \bar{R}\|_{L^2}$ at steady state.

This provides a functional inequality bound rather than a purely heuristic Gini claim.

9.6. Curvature and Power Singularity

Define local curvature:

$$\kappa(x) = -\Delta R(x). \quad (46)$$

From steady-state equation:

$$\kappa(x_0) = \frac{\gamma}{D(\rho(x_0))} R(x_0). \quad (47)$$

Thus concentration scales inversely with diffusion.

Density-dependent diffusion therefore directly controls curvature magnitude at innovation sites.

9.7. Interpretation

The continuous formulation establishes:

- Reward concentration arises from diffusiondecay equilibrium.
- Policy parameters (D_0, γ, α) tune curvature bounds.
- Inequality can be controlled via parabolic smoothing.

The Geozotic mechanism is therefore a tunable parabolic PDE with provable energy decay properties, not a heuristic redistribution metaphor.

*Reward diffuses under constraint;
inequality is curvature under control.*

10. Embodied Thermodynamics: Cooking, Play, and Skill Acquisition

*The test of a skill is not its
performance under guidance but
its transfer under surprise.*

The macro-thermodynamic claims of the preceding sections risk dismissal as speculative metaphor unless grounded in observable micro-scale phenomena. This section argues that everyday skill acquisition—exemplified by cooking, play, and mathematical abstraction—is the micro-scale instance of the same entropy-management logic that governs civilisational institutions.

10.1. Abstraction as Entropy Compression

A central claim is that skill consists in the capacity to perceive *ratios*, *substitutions*, and *invariants* rather than surface features. This maps directly onto the entropy efficiency index η_S (??).

Definition 10.1 (Compression Operator). An *abstraction* is a function $A : \mathcal{S} \rightarrow \mathcal{C}$ from a surface-feature space $\mathcal{S}$ to a compressed representation space $\mathcal{C}$, such that:

$$H(\mathcal{C}) \ll H(\mathcal{S}), \quad \text{and} \quad \Psi(A) \approx \Psi(\text{id}), \quad (48)$$

where H denotes Shannon entropy and Ψ measures task-relevant functional capacity.

A skilled cook who scales a recipe without measurement is operating in $\mathcal{C}$ (ratio space) rather than $\mathcal{S}$ (ingredient list space). The same compression underlies scaling laws in physics and type abstraction in programming. We call this capacity *structural literacy*.

Proposition 10.1 (High- η_S Cognition). Agents operating in $\mathcal{C}$ achieve higher η_S than those operating in $\mathcal{S}$, because:

$$\eta_S^{\mathcal{C}} = \frac{\Delta I}{\Delta E_{\mathcal{C}}} > \frac{\Delta I}{\Delta E_{\mathcal{S}}} = \eta_S^{\mathcal{S}}, \quad E_{\mathcal{C}} < E_{\mathcal{S}}. \quad (49)$$

Compression reduces the cognitive energy cost ΔE of generating equivalent informational output ΔI .

10.2. The Curry-Centric Hub: A Generative Grammar of Cuisine

Consider the claim that all cooked food can be reached from a canonical curry by incremental parameter modifications. This is more than metaphor: it describes a *transformation graph* over a continuous ingredient-parameter space.

Let $\mathcal{F}$ be the space of dishes, parameterised by vectors $\theta = (\text{fat}, \text{acid}, \text{umami}, \text{heat}, \text{sweetness}, \dots) \in \mathbb{R}^k$. A *culinary path* is a continuous curve $\gamma : [0, 1] \rightarrow \mathcal{F}$ such that $\gamma(0) = \theta_{\text{curry}}$ and $\gamma(1) = \theta_{\text{target}}$.

Culinary neoteny is the capacity to explore $\mathcal{F}$ safely within bounded constraints—analogueous to the explore/exploit balance of Gopnik’s developmental model. The skilled cook maintains a low-entropy internal map of $\mathcal{F}$ and can generate novel trajectories without consulting external recipes (tutorials).

10.3. Emulsification as Conceptual Integration

Emulsification—binding oil and water through an amphiphilic emulsifier (e.g. lecithin, mustard)—is a physical model of *conceptual binding*. Incompatible phases correspond to incommensurable domains of knowledge; emulsifiers correspond to bridging principles (mathematics, thermodynamics, grammar) that stabilise their conjunction.

$$\text{Concept}_A \oplus \text{Concept}_B \xrightarrow{\text{emulsifier principle}} \text{Stable hybrid concept}. \quad (50)$$

This directly parallels the institutional scaffolding of Section 2: the university is an emulsifier between curiosity (oil) and social utility (water).

10.4. Play as High-Variance Sampling

Definition 10.2 (Play). Play is high-variance sampling of action space under bounded cost:

$$\pi_{\text{play}}(a) \propto \exp(\beta \text{Var}(Q(s, a))), \quad \beta > 0, \quad (51)$$

where $Q(s, a)$ is the action-value function and β is an exploration temperature.

In reinforcement-learning terms, play increases exploration rate without catastrophic penalty, because the *cost of failure is subsidised*—by parents in childhood, by institutions in adulthood. Neotenous cognition is thus not sentimental but algorithmic: it is subsidised high- β sampling.

10.5. Boredom, Anxiety, and the Viable Entropy Band

Mental health, within this framework, corresponds to maintaining perceived entropy within a viable band $[\underline{S}, \overline{S}]$:

$$\underline{S} \leq S_{\text{perceived}}(t) \leq \overline{S}. \quad (52)$$

- **Boredom** occurs when $S_{\text{perceived}} < \underline{S}$: the environment offers fewer degrees of freedom than the agent’s capacity.
- **Anxiety** occurs when $S_{\text{perceived}} > \overline{S}$: uncertainty exceeds the agent’s predictive capacity.
- **Flow** occurs when $S_{\text{perceived}} \approx \frac{1}{2}(\underline{S} + \overline{S})$: challenge matches competence.

This formalises Csikszentmihalyi’s flow channel in entropic terms and connects to Friston’s free-energy principle (?): the agent minimises variational free energy by maintaining predicted entropy close to actual entropy.

Policy Implication

Cognitive sanctuaries are optimal if they keep participants within the viable entropy band (??): enough structure to prevent anxiety, enough freedom to prevent boredom. Calibrating institutional constraint density is therefore a thermodynamic design problem.

10.6. Skill Acquisition as Simulated Annealing

Iterative error-correction in learning—over-seasoning, burning, iterating—is literal simulated annealing. The learner begins at high temperature (wide exploration), identifies failure modes, and gradually cools toward stable heuristics.

$$T_{\text{skill}}(n) = T_0 e^{-\lambda n} + \varepsilon_{\text{play}}, \quad n = \text{number of practice episodes}, \quad (53)$$

where $\varepsilon_{\text{play}} > 0$ ensures that annealing never reaches absolute zero—preserving the capacity for continued exploration. This micro- scale schedule mirrors the civilisational annealing of Section ??.

10.7. Recipe Dependence Versus Structural Competence

Two learner types can be distinguished:

- **Surface-bound learners** follow tutorials without abstracting. They reduce local entropy (noise in a specific dish) but fail to extract invariants. η_S remains low because ΔE does not decrease across episodes.
- **Structural learners** abstract to $\mathcal{C}$ after a small number of episodes. η_S rises rapidly as compression yields generalisation.

Educational design can accelerate the transition from surface-bound to structural learning by presenting tasks that *require* ratio reasoning rather than memorisation—i.e. by engineering the zone of proximal development as an entropy gradient slightly above the learner’s baseline.

10.8. Autonomy, External Constraint, and Entropy Alignment

There is an apparent paradox in the observation that many highly autonomous individuals report preferring externally imposed constraints. The resolution is entropic: external constraints reduce *decision entropy* about goals, freeing cognitive bandwidth for execution. When goals are given, the agent need not sample from $\pi(\text{goals})$; the freed capacity is redirected toward the higher-resolution sampling of $\pi(\text{actions} \mid \text{goal})$.

This enriches the cognitive sanctuary concept: the institutional scaffold need not constrain thought—it need only constrain *goal selection*, leaving the exploration of means maximally free.

11. Abstraction, Kolmogorov Compression, and Transfer Stability

The entropy-efficiency index η_S introduced in (??) implicitly assumes that information gain ΔI reflects structured compression rather than raw accumulation. We formalise this claim using algorithmic information theory (??).

11.1. Kolmogorov Compression and Structural Literacy

Let x denote a task instance (e.g. a specific recipe, equation, or painting). Let $K(x)$ denote its Kolmogorov complexity: the length of the shortest program that generates x .

A learner operating at the surface level stores multiple instances $\{x_1, \dots, x_n\}$ independently, yielding total description length:

$$L_{\text{surface}} = \sum_{i=1}^n K(x_i). \quad (54)$$

A structural learner identifies a generating schema $\mathcal{G}$ such that:

$$x_i = \mathcal{G}(\theta_i), \quad (55)$$

for parameters θ_i in a lower-dimensional space. Total description length becomes:

$$L_{\text{structural}} = K(\mathcal{G}) + \sum_{i=1}^n K(\theta_i). \quad (56)$$

Proposition 11.1 (Compression Gain). If there exists $\mathcal{G}$ such that

$$K(\mathcal{G}) + \sum_{i=1}^n K(\theta_i) < \sum_{i=1}^n K(x_i), \quad (57)$$

then structural learning strictly reduces algorithmic description length.

This formalises the transition from instance-memorisation to generative reasoning. The “curry-centric hub” functions as $\mathcal{G}$: a compact program from which diverse dishes are instantiated by parameter variation.

11.2. Transfer Stability as Invariance Under Perturbation

Compression alone is insufficient; abstraction must also preserve performance under perturbation.

Let ϵ represent perturbation in input conditions (e.g. ingredient substitution, missing measurement, environmental noise). Let $\Psi(x)$ denote task performance.

Definition 11.1 (Transfer Stability). A representation is transfer-stable if there exists $C > 0$ such that:

$$|\Psi(x + \epsilon) - \Psi(x)| \leq C\|\epsilon\|, \quad (58)$$

uniformly over admissible task instances.

Surface-bound learners typically violate uniform Lipschitz continuity, whereas structural learners preserve invariants; perturbations act on parameters θ rather than the generating schema $\mathcal{G}$.

11.3. Predictive Processing and Energy Minimisation

Within the free-energy framework (?), an agent maintains a generative model $p_\theta(x)$ and updates parameters to minimise variational free energy:

$$F = \mathbb{E}_q[\log q(z) - \log p_\theta(x, z)]. \quad (59)$$

Optimal learning solves:

$$\min_{\theta} \left(\text{Prediction Error} + \beta \cdot \text{Model Complexity} \right), \quad (60)$$

where β controls the complexityaccuracy trade-off.

In reinforcement-learning language (?), β^{-1} corresponds to exploration temperature.

In embodied skill acquisition:

- Early learning: high temperature (broad hypothesis sampling).
- Later learning: annealed temperature (model consolidation).

This mirrors the annealing schedule of Section ??.

11.4. Structural Literacy as a Phase Transition

Model the transition from surface-bound to structural cognition as a representational phase transition.

Let H_n denote entropy of representation after n learning episodes. Empirically:

$$H_n = \begin{cases} H_0 - \delta n, & n < n_c, \\ H_{\text{schema}}, & n \geq n_c, \end{cases} \quad (61)$$

where n_c is a critical threshold at which the generative schema $\mathcal{G}$ is inferred.

The discontinuity at n_c corresponds to the abrupt reduction in effective description length – an informational analogue of a second-order phase transition.

11.5. Entropy Compression and Civilisational Scaling

At civilisational scale, innovation functions analogously. Discovery of generative schemas (e.g. calculus, electromagnetism, germ theory) reduces algorithmic description length across domains.

Define civilisational compression rate:

$$\mathcal{C}(t) := -\frac{d}{dt}(\text{Effective Description Length}). \quad (62)$$

The entropy-efficiency index η_S thus measures compression depth relative to energetic expenditure:

$$\eta_S \propto \frac{\mathcal{C}(t)}{\text{Energetic expenditure}}. \quad (63)$$

A steady-state civilisation maximises schema depth rather than material throughput.

To abstract is to lower entropy without lowering possibility.

12. The Geometry of Cuisine: A Manifold of Transformations

If abstraction compresses surface variability into generative schema, then cuisine provides a physically embodied example of such compression. We formalise the “curry-centric hub” as a differentiable manifold with metric structure, following standard geometric constructions for smooth fields and variational systems (?).

12.1. Dish Space as a Parameter Manifold

Let $\mathcal{F}$ denote the space of all possible dishes constructible from a finite ingredient basis $\mathcal{I}$ under admissible cooking operations $\mathcal{O}$ (heating, emulsifying, fermenting, etc.).

Each dish is represented as a parameter vector:

$$\theta = (\theta_1, \dots, \theta_k) \in \mathbb{R}^k, \quad (64)$$

where coordinates correspond to continuous flavour and structural axes: fat, acid, salt, heat, sweetness, umami, viscosity, protein density, carbohydrate density, water activity, etc.

Definition 12.1 (Cuisine Manifold). The cuisine manifold $(\mathcal{F}, g)$ is a smooth manifold equipped with Riemannian metric g_{ij} measuring perceptual distinguishability between dishes.

The metric tensor g_{ij} may be empirically estimated from psychophysical discrimination thresholds:

$$g_{ij} = \mathbb{E} \left[\frac{\partial^2 D(\theta, \theta')}{\partial \theta_i \partial \theta_j} \right], \quad (65)$$

where D measures perceptual distance. This induces a geometry analogous to configuration spaces in continuum mechanics (?).

12.2. Geodesics and Transformational Paths

A culinary transformation is a smooth curve $\gamma : [0, 1] \rightarrow \mathcal{F}$.

The geodesic equation:

$$\frac{d^2 \theta^k}{dt^2} + \Gamma_{ij}^k \frac{d\theta^i}{dt} \frac{d\theta^j}{dt} = 0, \quad (66)$$

describes minimal-effort transformations under metric g .

Interpretation: A skilled cook follows near-geodesic paths in flavour space, requiring minimal corrective energy. Novices traverse high-curvature trajectories, increasing energetic cost.

12.3. Connected Components and Cuisine Families

Define adjacency relation:

$$\theta \sim \theta' \quad \text{if} \quad d_g(\theta, \theta') < \epsilon, \quad (67)$$

for perceptual threshold ϵ .

Cuisine families correspond to connected components of $\mathcal{F}$ under this relation.

Proposition 12.1 (Connectivity of the Curry Hub). If θ_{curry} lies in the interior of $\mathcal{F}$ and all primary flavour axes remain nonzero, then the component containing θ_{curry} is dense in $\mathcal{F}$.

Interpretation: The curry parameter vector lies near a high-degree node in flavour space. Small perturbations generate soups, stews, sauces, pasta dishes, chili, shakshuka, mole, tagine, ramen, etc. Thus curry functions as a generative hub in $\mathcal{F}$.

12.4. Phase Boundaries

Certain culinary transitions involve discontinuities: emulsion breakage, protein coagulation, caramelisation thresholds. These correspond to structural instabilities analogous to phase transitions in physical systems (?).

Let $\Sigma \subset \mathcal{F}$ denote hypersurfaces where Jacobian determinant vanishes:

$$\det \left(\frac{\partial \mathcal{O}}{\partial \theta} \right) = 0. \quad (68)$$

Across Σ , small parameter changes yield large structural shifts. These are culinary phase transitions.

The emergence of structured dishes from local reaction-diffusion interactions (e.g., browning gradients, fermentation fronts) is formally analogous to morphogenetic pattern formation in chemical systems (?).

Skilled practitioners learn to operate near Σ without crossing destabilising boundaries.

12.5. Entropy of Cuisine

Define dish entropy as:

$$S(\theta) = - \sum_i p_i(\theta) \log p_i(\theta), \quad (69)$$

where p_i are perceptual or compositional proportions.

High entropy dishes lack structural coherence. Low entropy dishes may be bland. Optimal cuisine occupies an intermediate band:

$$S_{\min} < S(\theta) < S_{\max}. \quad (70)$$

This mirrors the viable entropy band (??) in cognitive and institutional systems.

12.6. Civilisational Analogy

Cultural systems resemble cuisine manifolds:

- Institutions are regions in parameter space.
- Revolutions are phase transitions across Σ .
- Traditions are local minima of action functionals.

The curry hub demonstrates that generativity emerges from central positioning within a richly connected manifold.

A cuisine is not a list of dishes but a geometry of possibility.

13. The Viable Entropy Band and Institutional Constraint Density

We formalise the Viable Entropy Band (??) as a controlled dynamical system in the sense of nonlinear stability analysis (?).

13.1. Perceived Entropy as a State Variable

Let $S_p(t)$ denote perceived environmental entropy (uncertainty, novelty, task variability), and let $C(t)$ denote institutional constraint density (rules, scaffolding, supervision, structure).

We model entropy dynamics as:

$$\frac{dS_p}{dt} = \alpha E(t) - \beta C(t) - \gamma A(t), \quad (71)$$

where:

- $E(t)$ = exogenous environmental variability,
- $C(t)$ = imposed constraint density,
- $A(t)$ = agent adaptation (skill acquisition),
- $\alpha, \beta, \gamma > 0$.

Exploration increases entropy. Structure reduces entropy. Adaptation internalises entropy, lowering perceived uncertainty. This internalisation corresponds to model updating under predictive free-energy minimisation (?).

13.2. Definition: Viable Entropy Band

Define bounds $\underline{S}, \bar{S}$ such that:

$$\underline{S} \leq S_p(t) \leq \bar{S}. \quad (72)$$

Empirically, performance and subjective engagement peak within an intermediate channel of challengeskill balance, as documented in flow studies (?).

- If $S_p < \underline{S}$ boredom regime.
- If $S_p > \bar{S}$ anxiety regime.
- If $S_p \in (\underline{S}, \bar{S})$ flow regime.

13.3. Equilibrium Constraint Density

At equilibrium $\frac{dS_p}{dt} = 0$:

$$C^* = \frac{\alpha E - \gamma A}{\beta}. \quad (73)$$

Thus optimal institutional constraint density depends on:

- Environmental volatility E ,
- Internal competence A .

As skill A increases, optimal constraint C^* decreases. Institutions that fail to relax structure as competence rises shift agents into the boredom regime.

13.4. Bifurcation Analysis

Let constraint density respond adaptively:

$$\frac{dC}{dt} = k(S_p - S_0), \quad (74)$$

where S_0 is target entropy midpoint.

Substituting (??) into (??) yields coupled system:

$$\frac{dS_p}{dt} = \alpha E - \beta C - \gamma A, \quad (75)$$

$$\frac{dC}{dt} = k(S_p - S_0). \quad (76)$$

Linearising around equilibrium (S_p^*, C^*) gives Jacobian:

$$J = \begin{pmatrix} 0 & -\beta \\ k & 0 \end{pmatrix}. \quad (77)$$

Eigenvalues satisfy:

$$\lambda^2 + \beta k = 0 \quad \Rightarrow \quad \lambda = \pm i\sqrt{\beta k}. \quad (78)$$

Thus the undamped system exhibits neutral oscillations, a centre in the phase plane (?).

Introduce damping term δS_p :

$$\frac{dS_p}{dt} = \alpha E - \beta C - \gamma A - \delta S_p. \quad (79)$$

Eigenvalues become:

$$\lambda = -\frac{\delta}{2} \pm \sqrt{\frac{\delta^2}{4} - \beta k}. \quad (80)$$

Stability condition:

$$\delta^2 > 4\beta k. \quad (81)$$

If violated, the system undergoes Hopf bifurcation, producing sustained oscillations in perceived entropy (chronic burnout/boredom cycling) (?).

13.5. Optimal Institutional Design

Define optimisation functional:

$$\mathcal{L}(C) = -(S_p - S_0)^2 - \lambda_C C^2. \quad (82)$$

Maximising $\mathcal{L}$ balances entropy deviation penalty against over-constraint cost.

First-order condition:

$$\frac{\partial \mathcal{L}}{\partial C} = 2\beta(S_p - S_0) - 2\lambda_C C = 0. \quad (83)$$

Yielding:

$$C^* = \frac{\beta}{\lambda_C}(S_p - S_0). \quad (84)$$

Thus optimal structure scales linearly with entropy deviation but is damped by institutional rigidity parameter λ_C .

13.6. Macro-Scaling to Civilisation

Replace S_p with collective cognitive entropy S_c , and C with governance intensity G :

$$\frac{dS_c}{dt} = \alpha E - \beta G - \gamma A_{\text{collective}}. \quad (85)$$

Too little governance yields instability. Excess governance suppresses adaptive internalisation, increasing long-run entropy through stagnation.

The steady-state civilisation operates at:

$$G^* = \frac{\alpha E - \gamma A}{\beta}, \quad (86)$$

keeping S_c within viable bounds while maintaining adaptive free-energy minimisation at collective scale (?).

*Stability is not equilibrium.
It is entropy regulated within viable bounds.*

14. Boredom and Anxiety as Spectral Phenomena in Predictive Dynamics

We sharpen the viable entropy band model by analysing the internal predictive dynamics of the agent. Rather than treating boredom and anxiety as scalar states, we derive them from spectral properties of the agents generative model using linear stability theory (?).

14.1. Generative Model as Linear Operator

Let the agent maintain a predictive model:

$$x_{t+1} = Ax_t + \xi_t, \quad (87)$$

where:

- $x_t \in \mathbb{R}^n$ is latent state estimate,
- A is internal transition operator,
- ξ_t is environmental perturbation (noise).

Prediction error:

$$e_t = x_{t+1}^{\text{obs}} - Ax_t. \quad (88)$$

Let Σ_ξ denote covariance of ξ_t .

14.2. Spectral Radius and Predictive Capacity

Let λ_i denote eigenvalues of A . The spectral radius:

$$\rho(A) = \max_i |\lambda_i|. \quad (89)$$

For linear discrete systems, stability requires $\rho(A) < 1$ (?).

- If $\rho(A) < 1$: trajectories contract to fixed point.
- If $\rho(A) = 1$: critical regime (marginal stability).
- If $\rho(A) > 1$: exponential divergence.

14.3. Boredom as Over-Contraction

Define environmental entropy rate:

$$H_E = \text{Ent}(\xi_t). \quad (90)$$

Define predictive entropy rate:

$$H_P = \text{Ent}(Ax_t). \quad (91)$$

Definition 14.1 (Boredom Condition). Boredom corresponds to the regime:

$$H_E \ll H_P \quad \text{and} \quad \rho(A) \ll 1. \quad (92)$$

In this regime, eigenmodes decay rapidly. State trajectories collapse to low-dimensional attractors. Environmental perturbations are insufficient to excite higher-order modes.

Spectrally, most eigenvalues lie deep within the unit circle, producing overdamped predictive dynamics.

14.4. Anxiety as Spectral Instability

Definition 14.2 (Anxiety Condition). Anxiety corresponds to:

$$H_E \gg H_P \quad \text{and/or} \quad \rho(A) \rightarrow 1^+. \quad (93)$$

Two mechanisms generate instability:

Environmental Overload:

$$\text{tr}(\Sigma_\xi) \gg \text{tr}(A\Sigma_x A^\top). \quad (94)$$

Internal Instability:

Largest eigenvalue exceeds unity, amplifying small prediction errors exponentially.

Anxiety corresponds to unstable eigenmodes dominating predictive dynamics.

14.5. Flow as Critical Spectral Balance

Definition 14.3 (Flow Condition). Flow occurs when:

$$H_E \approx H_P \quad \text{and} \quad \rho(A) \lesssim 1. \quad (95)$$

At this boundary:

- Prediction errors excite multiple eigenmodes.
- None diverge.
- Learning gradients remain high.

Empirically, biological and neural systems exhibit signatures of operation near critical points, maximising dynamic range and information transmission near $\rho(A) \approx 1$ (??).

Thus flow corresponds to near-critical predictive dynamics.

14.6. Lyapunov Function for Affective Stability

Define Lyapunov candidate:

$$V(x) = x^\top P x, \quad (96)$$

with P positive definite.

System stable if:

$$A^\top P A - P < 0, \quad (97)$$

which is equivalent to $\rho(A) < 1$ for some $P > 0$ (?).

Institutional scaffolding modifies dynamics:

$$A' = A - \Gamma, \quad (98)$$

where Γ is a constraint operator.

Optimal scaffolding sets:

$$|\lambda_i(A')| \approx 1 - \epsilon, \quad 0 < \epsilon \ll 1. \quad (99)$$

Excessive damping ($\epsilon \gg 0$) pushes system into boredom regime. Insufficient damping permits instability.

14.7. Entropy, Affect, and Civilisation

At civilisational scale, replace A with institutional transition operator governing idea propagation.

- Over-constrained systems: $\rho(A) \ll 1$ (overdamped stagnation).
- Unregulated amplification: $\rho(A) > 1$ (runaway instability).
- Stable innovation: $\rho(A) \approx 1$ (critical regime).

Criticality without divergence is therefore a unifying stability condition across individual cognition and collective systems.

Boredom is overdamped prediction.

Anxiety is unstable amplification.

Flow is critical balance.

15. Collective Autocatalysis and the Reversal of Selection

Darwinian frameworks often treat evolution as a contest among selfish replicators. Yet such models omit the thermodynamic substrate on which replication occurs. When energy gradients are steep and resources scarce, competition dominates; as entropy differentials are smoothed, cooperation becomes the lower-energy path.

15.1. Beyond the Selfish Gene

The “selfish gene” metaphor mistakes a bookkeeping identity for a causal principle. Genes do not act; they persist within autocatalytic environments of mutual reinforcement. The stability of any replicator depends on the collective coherence of its host system. What appears as competition at one scale often manifests as autocatalysis at another: networks of entities that reproduce one another’s conditions of viability (??).

$$\frac{dN_i}{dt} = \sum_j k_{ij}N_j - \lambda_i N_i, \quad k_{ij} > 0 \implies \text{mutual reinforcement.} \quad (100)$$

When the coupling matrix $\mathbf{K} = (k_{ij})$ is symmetric and positive semi-definite, the system converges toward collective persistence rather than zero-sum exclusion.

15.2. Autocatalytic Sets and Social Equilibria

Human societies behave as autocatalytic ensembles. Each act of maintenance or empathy reduces local uncertainty, allowing others to redirect energy from defence to creation. Lowering entropy locally does not invite exploitation; it raises the carrying capacity of trust.

$$\frac{dS_{\text{local}}}{dt} < 0 \implies \frac{dR_{\text{collective}}}{dt} > 0, \quad (101)$$

where $R_{\text{collective}}$ denotes the group's resilience. Empirically, cooperation and altruism increase when environmental stress decreases (?).

15.3. Asymmetry and Breakdown

Real-world coupling matrices include negative entries ($k_{ij} < 0$), representing competition, exploitation, or misinformation. Such asymmetries can destabilise autocatalytic sets, driving systems toward exclusionary equilibria. Steady-state governance must actively damp negative feedbacks through resource redistribution, informational transparency, or normative sanctions.

15.4. The Entropic Basis of Safety and Affection

Sexual-selection narratives often emphasise dominance and scarcity—transient phenomena of high-entropy regimes. In cooperative niches, attraction shifts toward indicators of stability and care. Affection can be interpreted as an entropy-minimising feedback: to be near the calm is to lower one's thermodynamic cost of prediction. Group cohesion is rewarded not only culturally but biophysically; safety propagates as a contagious gradient.

15.5. Selection as Energy Accounting

$$\frac{dW}{dt} = -T \frac{dS_{\text{collective}}}{dt}, \quad (102)$$

where W is adaptive work and T is the effective temperature of social tension. Evolution proceeds by minimising T while maintaining $W > 0$ —a process identical in form to the thermodynamic optimisation of life itself (??).

15.6. The Cooperative Minimum

The steady-state civilisation, viewed through this lens, is an attractor of cooperative minima. Competition remains, but as modulation, not motive.

*Every act that lowers entropy for others lowers the danger for oneself.
The unconscious reward for collective safety is calm; its conscious expression is compassion.*

16. Annealing, Ageing, and the Intelligence of Care

If neoteny preserves juvenile exploration within adulthood, ageing redistributes it across generations. The “grandmother effect” in evolutionary anthropology suggests that post-reproductive individuals increase group fitness by investing in others’ offspring. This implies a second-order developmental logic: intelligence is distributed across time within lineages.

16.1. The Grandparent Problem

Nature appears to maintain a collective form of neoteny across individuals. Elders shift from exploitation to empowerment. Empirically, ageing correlates with rising altruism and subjective well-being despite physical decline—“hedonic inversion” (?).

$$\frac{dS_{\text{self}}}{dt} > 0 \implies \frac{dS_{\text{collective}}}{dt} < 0. \quad (103)$$

By accepting local entropy (physical decline), elders lower systemic tension and permit wider exploration by others. The *grandparent hypothesis* holds that longevity beyond fertility evolves because it stabilises the developmental niche of the young.

16.2. Simulated Annealing as a Model of Ageing

$$T_{\text{cognitive}}(t) = T_0 e^{-\lambda t} + \varepsilon_{\text{elder}}, \quad (104)$$

where $\varepsilon_{\text{elder}}$ represents the late-life cognitive reheating that prevents premature convergence of cultural norms. Each generation provides a new cooling phase; elders rehearse small reheatings that maintain flexibility. Less top-down control—

often pathologised as decline—may reopen the search space of ideas and empathic responses.

16.3. Empirical Correlates of Cognitive Cooling

Neuroimaging meta-analyses reveal reduced prefrontal cortical thickness and diminished executive control in older adults, accompanied by heightened default-mode network variability—patterns consistent with increased exploratory sampling (?). Longitudinal well-being studies document U-shaped happiness trajectories, with late-life peaks attributable to socioemotional selectivity and reduced future discounting.

16.4. The Care-Alignment Analogy

Every elder faces a version of the alignment problem: how to transmit goals and values that remain adaptive under novel conditions? The caregiving niche solves this by coupling tradition and innovation—elders supply scaffolds of safety within which the young can diverge. The function of care is not obedience but *bounded freedom*.

In artificial intelligence, simulated annealing is likewise used to navigate rugged optimisation landscapes. Nature, however, has perfected it: ageing constitutes repeated rounds of annealing, with care as the cooling mechanism that consolidates learning without freezing novelty.

16.5. Entropy, Safety, and Happiness

$$\Delta S_{\text{group}} < 0 \iff \Delta U_{\text{individual}} > 0. \quad (105)$$

The steady-state civilisation will depend on this principle: its elders—human or artificial—serve as dynamic annealers, reintroducing flexibility and empathy whenever systems threaten to over-fit their own success.

Care is not sentiment but architecture.

Coda. The intelligence of care is the long-term cooling of civilisation. It tempers the heat of innovation with the calm of understanding. Each generation warms, explores, and cools again—not to extinguish its flame, but to prevent the fire from consuming the hearth.

17. Multi-Scale Entropy Dynamics: From Skill Annealing to Civilisational Fields

We unify micro-dynamics of skill acquisition with macro-dynamics of civilisational field structure. The framework follows multi-scale dynamical aggregation and renormalisation principles (??), with interpretive grounding in multi-scale predictive inference (?).

17.1. Micro-Level: Skill Annealing

Individual exploratory temperature evolves as:

$$T_i(t) = T_{i,0}e^{-\lambda_i t} + \varepsilon_i, \quad (106)$$

where:

- T_i = exploratory temperature of agent i ,
- λ_i = learning rate,
- ε_i = residual exploratory variance.

Define entropy contribution:

$$S_i(t) = c_T T_i(t), \quad (107)$$

with proportionality constant $c_T > 0$.

Exploration injects entropy; consolidation lowers it.

17.2. Mesoscopic Aggregation

Let agents occupy positions $x_i(t)$. Define entropy density:

$$S(x, t) = \sum_i S_i(t) \delta(x - x_i(t)). \quad (108)$$

Coarse-graining over neighbourhood $\mathcal{B}_\ell(x)$:

$$S_\ell(x, t) = \frac{1}{|\mathcal{B}_\ell|} \int_{\mathcal{B}_\ell(x)} S(y, t) dy. \quad (109)$$

Assume effective mesoscopic dynamics:

$$\frac{\partial S}{\partial t} = -\Lambda(x)S + \Xi(x) + D_S \Delta S, \quad (110)$$

where:

- $\Lambda(x)$ = institutional cooling rate,
- $\Xi(x)$ = innovation injection rate,
- D_S = entropy diffusion coefficient.

Institutions thus appear as spatially varying annealing operators.

17.3. Coupling to RSVP Field

Let $\Phi(x, t)$ denote constraint potential field.

We couple:

$$\frac{\partial \Phi}{\partial t} = \alpha S - \beta \Phi + D_\Phi \Delta \Phi. \quad (111)$$

This produces a coupled linear reaction-diffusion system, consistent with the master operator in Appendix ??.

17.4. Emergent Steady-State Condition

At spatially homogeneous equilibrium:

$$0 = -\Lambda S^* + \Xi, \quad (112)$$

$$0 = \alpha S^* - \beta \Phi^*. \quad (113)$$

Hence:

$$S^* = \frac{\Xi}{\Lambda}, \quad \Phi^* = \frac{\alpha}{\beta} \frac{\Xi}{\Lambda}. \quad (114)$$

Potential scales linearly with innovation, inversely with institutional cooling.

17.5. Flow Field Emergence

Define agency field:

$$v = -\nabla \Phi. \quad (115)$$

Thus steady-state flow:

$$v^* = -\nabla \left(\frac{\alpha}{\beta} \frac{\Xi}{\Lambda} \right). \quad (116)$$

Spatial gradients in innovation-to-cooling ratio generate macroscopic flow.

17.6. Scaling Law

Let average learning rate:

$$\bar{\lambda}(x) = \frac{1}{N_x} \sum_{i \in \mathcal{B}(x)} \lambda_i. \quad (117)$$

Assume institutional cooling depends on competence:

$$\Lambda(x) = \Lambda_0 + \kappa \bar{\lambda}(x). \quad (118)$$

Then:

$$\Phi^* \propto \frac{\Xi}{\Lambda_0 + \kappa \bar{\lambda}}. \quad (119)$$

Regions with higher competence self-dampen entropy, approaching lower-temperature steady states.

17.7. Renormalisation Perspective

Define coarse-graining operator $\mathcal{R}_\ell$:

$$\mathcal{R}_\ell S(x) = b^{-d} \int_{\mathcal{B}_\ell(x)} S(y) dy, \quad (120)$$

with scaling factor $b > 1$, dimension d .

Assume scaling law:

$$S_{\ell+1} = b^{-z} S_\ell, \quad (121)$$

where z is entropy scaling exponent.

Under renormalisation group flow (?), fixed point satisfies:

$$\mathcal{R}_\ell(S^*, \Phi^*, v^*) = (S^*, \Phi^*, v^*). \quad (122)$$

Near criticality, observables obey power-law scaling with universal exponents (?).

Thus civilisational stability corresponds to RG fixed point balancing innovation injection and cooling dissipation.

17.8. Unified Principle

Micro skill annealing reduces entropy locally. Aggregated spatially, entropy evolves under reaction-diffusion dynamics. Coupling generates potential fields. Gradients drive flow. Renormalisation governs scale invariance.

Multi-scale inference models similarly describe hierarchical error minimisation across levels of organisation (?).

Civilisation is annealing across scales.

18. A Lagrangian Formulation of Civilisation in RSVP-Space

The RSVP identity (??) asserts conservation of a composite quantity:

$$\Phi + \chi(v) + S = \mathcal{C}, \quad (123)$$

where Φ denotes stored potential, v denotes agency flow, S denotes entropy density, and $\mathcal{C}$ is an invariant.

We treat civilisation as a field theory over spacetime $\Omega \subset \mathbb{R}^{3+1}$, following standard variational PDE constructions (?).

18.1. Field Variables

Let

$$\Phi(x, t), \quad v_i(x, t), \quad S(x, t) \quad (124)$$

be sufficiently smooth fields.

Define Lagrangian density:

$$\mathcal{L} = \frac{1}{2}\rho_v|v|^2 - \frac{1}{2}\kappa_\Phi|\nabla\Phi|^2 - U(\Phi, S) - \lambda(x, t)(\Phi + S + \chi(v) - \mathcal{C}), \quad (125)$$

where λ is a spacetime-dependent Lagrange multiplier.

18.2. Variational Principle

The action is:

$$\mathcal{S} = \int_{\Omega} \mathcal{L} d^3x dt. \quad (126)$$

Stationarity under variations of v , Φ , S , and λ yields EulerLagrange equations (?).

18.3. Flow Equation

Variation with respect to v gives:

$$\rho_v v_i - \lambda \frac{\partial \chi}{\partial v_i} = 0. \quad (127)$$

If $\chi(v) = \frac{1}{2}|v|^2$, then:

$$(\rho_v - \lambda)v = 0. \quad (128)$$

Thus either $v = 0$ or $\lambda = \rho_v$. Flow magnitude is therefore determined by constraint field.

18.4. Potential Equation

Variation with respect to Φ :

$$-\kappa_\Phi \Delta \Phi - \frac{\partial U}{\partial \Phi} - \lambda = 0. \quad (129)$$

18.5. Entropy Equation

Variation with respect to S :

$$-\frac{\partial U}{\partial S} - \lambda = 0. \quad (130)$$

18.6. Constraint Equation

Variation with respect to λ enforces:

$$\Phi + S + \chi(v) = \mathcal{C}. \quad (131)$$

18.7. Choice of Interaction Potential

Let

$$U(\Phi, S) = \alpha \Phi S + \frac{\beta}{2} S^2. \quad (132)$$

Then:

$$\frac{\partial U}{\partial \Phi} = \alpha S, \quad (133)$$

$$\frac{\partial U}{\partial S} = \alpha \Phi + \beta S. \quad (134)$$

From (??):

$$\lambda = -(\alpha \Phi + \beta S). \quad (135)$$

Substitute into (??):

$$-\kappa_\Phi \Delta \Phi - \alpha S + \alpha \Phi + \beta S = 0. \quad (136)$$

Rearranging:

$$\kappa_\Phi \Delta \Phi = \alpha(\Phi - S) + \beta S. \quad (137)$$

This yields a coupled elliptic system.

18.8. Steady-State Solution

In homogeneous steady state:

$$\Delta \Phi = 0, \quad \nabla \lambda = 0. \quad (138)$$

Thus:

$$\Phi^* + S^* + \chi(v^*) = \mathcal{C}. \quad (139)$$

Uniform solutions satisfy algebraic balance between potential, entropy, and flow.

18.9. Linear Stability

Perturb:

$$\Phi = \Phi^* + \delta \Phi, \quad S = S^* + \delta S. \quad (140)$$

Linearised operator has form:

$$\kappa_\Phi \Delta \delta \Phi = \alpha(\delta \Phi - \delta S) + \beta \delta S. \quad (141)$$

Instability occurs when effective mass term becomes negative:

$$\alpha - \beta < 0. \quad (142)$$

This corresponds to entropy dominating potential coupling, producing amplification of spatial modes.

18.10. Thermodynamic Analogy

The constraint multiplier λ acts analogously to temperature or chemical potential. Jacobson's derivation of Einstein equations from thermodynamic constraints (?) demonstrates that conservation laws can generate effective field equations from entropic principles.

Here, the RSVP constraint plays a comparable structural role: field equations arise from conservation of a composite invariant.

18.11. Geodesics in RSVP-Space

Define metric on configuration space:

$$ds^2 = a d\Phi^2 + b dv^2 + c dS^2. \quad (143)$$

The constraint surface:

$$\Phi + S + \chi(v) = \mathcal{C} \quad (144)$$

defines a codimension-1 hypersurface. Long-term evolution corresponds to extremal paths restricted to this hypersurface.

Constrained fields define structured motion.

18.12. Kinetic Terms and a Hyperbolic RSVP System

To obtain genuine time dynamics (rather than a purely elliptic constraint system), include kinetic terms for Φ and S in the Lagrangian density. Let $\alpha_\Phi, \alpha_S > 0$ be inertial coefficients. Define

$$\mathcal{L} = \frac{1}{2}\alpha_\Phi(\partial_t\Phi)^2 + \frac{1}{2}\alpha_S(\partial_t S)^2 + \frac{1}{2}\rho_v|v|^2 - \frac{1}{2}\kappa_\Phi|\nabla\Phi|^2 - \frac{1}{2}\kappa_S|\nabla S|^2 - U(\Phi, S) - \lambda(x, t)(\Phi + S + \chi(v) - \mathcal{C}), \quad (145)$$

with $\kappa_S > 0$ an entropy-stiffness (smoothing) parameter. Assume sufficiently regular fields and (for energy identities) either periodic boundary conditions or zero-flux conditions $\nabla\Phi \cdot n = \nabla S \cdot n = 0$ on $\partial\Omega$.

Euler–Lagrange equations. Stationarity of the action $\int_{\Omega} \mathcal{L} d^3x dt$ yields, for Φ and S ,

$$\alpha_{\Phi} \partial_{tt} \Phi - \kappa_{\Phi} \Delta \Phi + \frac{\partial U}{\partial \Phi}(\Phi, S) + \lambda = 0, \quad (146)$$

$$\alpha_S \partial_{tt} S - \kappa_S \Delta S + \frac{\partial U}{\partial S}(\Phi, S) + \lambda = 0. \quad (147)$$

Variation with respect to λ enforces the RSVP constraint

$$\Phi + S + \chi(v) = \mathcal{C}. \quad (148)$$

Variation with respect to v gives the algebraic flow relation

$$\rho_v v_i - \lambda \frac{\partial \chi}{\partial v_i}(v) = 0. \quad (149)$$

A convenient choice is $\chi(v) = \frac{1}{2}|v|^2$, which implies $\rho_v v - \lambda v = 0$ and therefore either $v = 0$ or $\lambda = \rho_v$ on the support of v .

Explicit coupling example. For

$$U(\Phi, S) = \alpha \Phi S + \frac{\beta}{2} S^2 + \frac{\eta}{2} \Phi^2, \quad (150)$$

one has $\partial_{\Phi} U = \alpha S + \eta \Phi$ and $\partial_S U = \alpha \Phi + \beta S$, so (??)–(??) become

$$\alpha_{\Phi} \partial_{tt} \Phi - \kappa_{\Phi} \Delta \Phi + \eta \Phi + \alpha S + \lambda = 0, \quad (151)$$

$$\alpha_S \partial_{tt} S - \kappa_S \Delta S + \alpha \Phi + \beta S + \lambda = 0. \quad (152)$$

Together with (??) and (??), this is a *constrained hyperbolic (wave-like) system* with spatial smoothing.

18.13. Hamiltonian Formulation and Conserved Energy

Define canonical momenta

$$\Pi_{\Phi} := \frac{\partial \mathcal{L}}{\partial(\partial_t \Phi)} = \alpha_{\Phi} \partial_t \Phi, \quad \Pi_S := \frac{\partial \mathcal{L}}{\partial(\partial_t S)} = \alpha_S \partial_t S. \quad (153)$$

The Hamiltonian density is

$$\mathcal{H} := \Pi_{\Phi} \partial_t \Phi + \Pi_S \partial_t S - \mathcal{L}. \quad (154)$$

Substituting (??) yields

$$\mathcal{H} = \frac{1}{2\alpha_\Phi} \Pi_\Phi^2 + \frac{1}{2\alpha_S} \Pi_S^2 + \frac{1}{2} \kappa_\Phi |\nabla \Phi|^2 + \frac{1}{2} \kappa_S |\nabla S|^2 + U(\Phi, S) + \lambda(\Phi + S + \chi(v) - \mathcal{C}) - \frac{1}{2} \rho_v |v|^2. \quad (155)$$

The total energy functional is

$$\mathcal{E}(t) = \int_\Omega \mathcal{H}(x, t) dx. \quad (156)$$

Under boundary conditions that eliminate flux terms (periodic or Neumann) and assuming λ imposes a time-independent constraint target $\mathcal{C}$, one obtains the formal conservation law

$$\frac{d}{dt} \mathcal{E}(t) = 0, \quad (157)$$

for the undamped system. If later you add dissipation terms (e.g. $-\delta_\Phi \partial_t \Phi$ and $-\delta_S \partial_t S$), then $\frac{d}{dt} \mathcal{E}(t) \leq 0$ becomes a Lyapunov decay statement.

18.14. Dispersion Relation for Linearised Perturbations

Linearise around a homogeneous equilibrium $(\Phi^*, S^*, v^*, \lambda^*)$ with constant coefficients. Write $\Phi = \Phi^* + \varphi$, $S = S^* + s$, $\lambda = \lambda^* + \ell$ and retain first order terms. For the quadratic potential (??), the linearised PDEs are

$$\alpha_\Phi \partial_{tt} \varphi - \kappa_\Phi \Delta \varphi + \eta \varphi + \alpha s + \ell = 0, \quad (158)$$

$$\alpha_S \partial_{tt} s - \kappa_S \Delta s + \alpha \varphi + \beta s + \ell = 0. \quad (159)$$

If v is held fixed in the background (or $v^* = 0$), the linearised constraint reduces to

$$\varphi + s = 0, \quad (160)$$

so $s = -\varphi$. Substituting into (??) gives

$$\alpha_\Phi \partial_{tt} \varphi - \kappa_\Phi \Delta \varphi + (\eta - \alpha) \varphi + \ell = 0, \quad (161)$$

and substituting into (??) gives

$$-\alpha_S \partial_{tt} \varphi + \kappa_S \Delta \varphi + (\alpha - \beta) \varphi + \ell = 0. \quad (162)$$

Eliminating ℓ yields a single constrained wave equation for φ :

$$(\alpha_\Phi + \alpha_S)\partial_{tt}\varphi - (\kappa_\Phi + \kappa_S)\Delta\varphi + (\eta - \beta)\varphi = 0. \quad (163)$$

Now take plane-wave ansatz $\varphi(x, t) = \hat{\varphi} e^{i(k \cdot x - \omega t)}$. Then $\Delta\varphi = -|k|^2\varphi$ and (??) gives the dispersion relation

$$\omega^2(k) = \frac{(\kappa_\Phi + \kappa_S)|k|^2 + (\eta - \beta)}{\alpha_\Phi + \alpha_S}. \quad (164)$$

Stability criterion. Linear stability requires $\omega^2(k) \geq 0$ for all k , which holds if and only if

$$\kappa_\Phi + \kappa_S > 0 \quad \text{and} \quad \eta - \beta \geq 0. \quad (165)$$

If $\eta - \beta < 0$, then low- $|k|$ modes satisfy $\omega^2(k) < 0$ and grow exponentially (a long-wavelength instability). The critical wavenumber where ω^2 changes sign is

$$|k_c|^2 = \frac{\beta - \eta}{\kappa_\Phi + \kappa_S}, \quad (\beta > \eta), \quad (166)$$

so only modes with $|k| < |k_c|$ are unstable.

Interpretation. The combined stiffness $(\kappa_\Phi + \kappa_S)$ suppresses short-wavelength structure, while the effective mass $(\eta - \beta)$ controls whether the homogeneous state is stable or undergoes large-scale symmetry breaking.

*The dynamical RSVP system behaves as a constrained coupled wave field;
its stability is decided by a mass term and a stiffness term.*

19. The Master Equation of Civilisational Entropy

We now synthesise the preceding results into a single dynamical system governing civilisation as a constrained entropy field. The formulation combines reaction-diffusion dynamics, advective transport, and institutional annealing within a coupled PDE framework in the sense of classical parabolic systems (?). The interpretation is compatible with predictive-regulatory dynamics in adaptive systems (?) and with thermodynamic approaches to economic organisation (??).

19.1. State Variables

Let $\Phi(x, t)$ denote structural potential (stored capacity), $v(x, t)$ denote agency flow, $S(x, t)$ denote relational entropy, $R(x, t)$ denote the reward field, and $\rho(x, t)$ denote population density, all defined on a domain $\Omega \subset \mathbb{R}^n$.

19.2. Core Coupled System

The full multi-scale model reduces to the coupled system

$$\frac{\partial S}{\partial t} = \Xi(x, t) - \Lambda(x, t)S - \nabla \cdot (vS), \quad (167)$$

$$\frac{\partial \Phi}{\partial t} = \alpha S - \beta \Phi + D_\Phi \Delta \Phi, \quad (168)$$

$$\frac{\partial R}{\partial t} = \nabla \cdot (D(\rho) \nabla R) - \gamma R + \sigma \Phi, \quad (169)$$

$$v = -\nabla \Phi, \quad (170)$$

together with the RSVP constraint

$$\Phi + \chi(v) + S = \mathcal{C}. \quad (171)$$

The entropy equation (??) is a transportreaction equation: innovation Ξ injects entropy, institutional cooling Λ removes it, and agency advects it across space. The potential equation (??) is a linear reactiondiffusion system in which entropy acts as a source term. The reward equation (??) is a density-corrected diffusion process coupled to structural capacity. Equation (??) closes the system by identifying agency with the gradient flow of potential, making the dynamics variational in spirit.

19.3. Master Efficiency Functional

Define the global entropy-efficiency functional

$$\tilde{\eta}_S(t) = \frac{\int_\Omega \alpha S(x, t) dx}{\int_\Omega \Xi(x, t) dx} - \kappa \int_\Omega |\nabla R|^2 dx. \quad (172)$$

The first term measures entropy converted into structured potential per unit innovation input, while the second penalises curvature concentration in the reward field. The curvature penalty reflects the thermodynamic cost of extreme concentration in spatial economic systems (??). In steady state the civilisation maximises $\tilde{\eta}_S$ subject to the invariant constraint (??).

Conservation of the composite invariant implies

$$\frac{d}{dt} \int_{\Omega} (\Phi + S) dx = 0, \quad (173)$$

which expresses dynamic redistribution rather than net expansion. This is structurally analogous to conservation laws in parabolicelliptic PDE systems (?) and to constrained free-energy minimisation in adaptive inference models (?).

19.4. Steady-State Solution

At equilibrium the reaction terms balance:

$$S^* = \frac{\Xi}{\Lambda}, \quad (174)$$

$$\Phi^* = \frac{\alpha}{\beta} S^*, \quad (175)$$

$$\Delta \Phi^* = 0, \quad (176)$$

and the reward field relaxes toward minimal curvature subject to boundary conditions. The steady-state therefore represents a regulated reactiondiffusionannealing system whose macroscopic order is sustained by continual entropy throughput rather than by static accumulation.

*Civilisation is entropy regulated by care,
potential generated by exploration,
and flow guided by structure.*

20. The Entropy Law in Bioeconomics

Nicholas Georgescu-Roegen's integration of the second law of thermodynamics into economic theory provides the rigorous physical foundation for the entropic framework developed throughout this essay (?).

$$\Delta S \geq 0, \quad (177)$$

with equality only in idealised reversible processes.

20.1. Irreversibility and Resource Degradation

Economic processes dissipate low-entropy inputs E_L into useful work E_U and high-entropy outputs E_H , satisfying $E_L = E_U + E_H$. Entropy increase follows from the

Clausius inequality:

$$\Delta S = S_H - S_L \geq \int \frac{\delta Q}{T}. \quad (178)$$

20.2. Bioeconomic Production Function

Georgescu-Roegen proposes a thermodynamically constrained production function:

$$Q = f(L, K, R; S), \quad (179)$$

where L is labour, K capital, R low-entropy resources, and S the entropic constraint. Growth accelerates $\dot{S}$, depleting finite low-entropy stocks.

20.3. Critique of Neoclassical Economics

Georgescu-Roegen's most incisive contribution demolishes neoclassical orthodoxy, which treats the economy as a closed, reversible system. Three myths are identified:

- **Perfect substitutability:** Capital cannot replace resources indefinitely; substitution occurs only within bounds set by the entropy law.
- **Circular flow fallacy:** Every production cycle leaves high-entropy residue; the pendulum model omits irreversibility.
- **Discounting the future:** Exponential discounting accelerates entropic debt across generations.

20.4. Implications for Steady-State Design

In the present framework, institutions function as entropy converters. The steady-state equilibrium $\dot{S}_{\text{usable}} = 0$ requires balancing exploratory dissipation with restorative care. Georgescu-Roegen's bioeconomics supplies the physical boundary condition for a civilisation that sustains neotenous cognition indefinitely.

21. Empirical Falsifiability and Testable Predictions

A framework is scientific only if it generates disconfirmable quantitative predictions. We therefore derive operational tests of the master system (??)(??).

21.1. Prediction 1: Quadratic InequalityEfficiency Law

From (??):

$$\eta_S = \eta_0 - \kappa_3 G^2 + \varepsilon, \quad (180)$$

where:

- η_S = innovation efficiency (patents or output per R&D unit),
- G = Gini coefficient of innovation reward distribution,
- ε = residual noise.

Test design: Panel regression across regions or firms.

$$\eta_{it} = \beta_0 + \beta_1 G_{it} + \beta_2 G_{it}^2 + u_{it}. \quad (181)$$

Prediction: $\beta_2 < 0$.

This aligns with endogenous growth theory debates in innovation economics (?) and inequality measurement literature (??).

Falsification: If $\beta_2 \geq 0$ consistently across robust specifications, the curvature penalty assumption fails.

21.2. Prediction 2: EntropyConstraint Stability Threshold

From Section ??, stability requires:

$$\delta^2 > 4\beta k. \quad (182)$$

Operationally:

- δ institutional damping index
- βk feedback gain between entropy deviation and regulation

Prediction: Burnoutboredom oscillation frequency increases as empirical proxy of $(4\beta k - \delta^2)$ becomes positive.

Test: Time-series spectral analysis of workforce turnover, absenteeism, and productivity volatility.

Falsification: If no oscillatory amplification occurs in underdamped regimes, the viable entropy band model fails.

21.3. Prediction 3: Spectral Criticality and Flow

From Section ??:

$$\rho(A) \approx 1 \iff \text{flowstates}. \quad (183)$$

Operationalisation: Estimate largest Lyapunov exponent or spectral radius from neural time-series (EEG/MEG/fMRI).

Prediction: Reported flow states correspond to near-critical neural avalanche statistics or power-law scaling (??).

Falsification: If flow corresponds instead to strongly subcritical or supercritical regimes, the spectral model fails.

21.4. Prediction 4: Density-Corrected Diffusion Reduces Dispersion

From Section ??, diffusion energy inequality implies:

$$\|R - \bar{R}\|_{L^2} = f(\alpha), \quad \frac{df}{d\alpha} < 0. \quad (184)$$

Operational test: Policy pilots implementing density-corrected reward redistribution.

Measure spatial dispersion before/after reform.

Prediction: Spatial variance of rewards decreases monotonically with diffusion exponent α .

Falsification: If dispersion remains unchanged or increases, the diffusion control mechanism fails.

21.5. Prediction 5: InnovationCooling Scaling Law

From (??):

$$\Phi^* \propto \frac{\Xi}{\Lambda}. \quad (185)$$

Operational proxies:

- Ξ innovation injection rate
- Λ institutional rigidity (tenure rigidity, rule density)

Prediction: Innovation output scales inversely with institutional overcooling, controlling for resource input.

Falsification: If highly rigid environments sustain high long-term innovation per resource unit, the multi-scale coupling fails.

21.6. Prediction 6: Renormalisation Fixed Point

The RG condition:

$$\mathcal{R}_\ell(S, \Phi, v) = (S, \Phi, v) \quad (186)$$

predicts scale-invariant statistics in steady-state regimes.

Test: Examine whether innovation variance, inequality dispersion, and productivity fluctuations exhibit power-law scaling with stable exponents (?).

Falsification: Persistent exponential divergence without stabilising exponents contradicts the renormalisation hypothesis.

21.7. Strong Falsification Criterion

The framework is falsified if robust longitudinal evidence shows:

$$\exists \text{ stable regime where } G \gg 0 \wedge \frac{\partial \eta_S}{\partial G^2} \geq 0 \text{ persistently.} \quad (187)$$

That is: high inequality coexists with sustained high innovation efficiency without diffusion correction.

21.8. Epistemic Status

The framework generates parameterised, regressible relationships between inequality and innovation efficiency, specifies spectral signatures in neural dynamics associated with boredom, anxiety, and flow, produces reaction-diffusion dispersion bounds governing reward curvature, and yields renormalisation-scale fixed-point conditions testable through longitudinal macroeconomic data. These claims are not metaphorical but operational: each corresponds to measurable quantities that can be subjected to statistical, spectral, or PDE-based evaluation.

The framework is therefore empirically vulnerable along multiple independent axes. It can fail through regression analysis, through neural criticality measurements, through spatial diffusion experiments, or through scale aggregation tests. Its scientific status rests precisely on this multi-domain exposure to refutation.

*A model earns credibility not by elegance,
but by surviving attempted refutation.*

22. Conclusion: The Thermodynamics of Care

Across this essay, a single theme has emerged: intelligence is not a solitary contest but a collective negotiation with entropy. From the neoteny of the child to the altruism of the elder, from the curiosity of the researcher to the compassion of the caregiver, the same dynamic repeats: local disorder is tolerated so that global coherence may increase.

22.1. From Genes to Civilisations

The traditional narrative of evolution captures only the early, high-temperature phase of biological history. As complexity rises, replication gives way to autocatalysis; selection shifts from the survival of the fittest individual to the persistence of the most stable network.

22.2. The Architecture of Equilibrium

Institutions, economies, and technologies are successful not when they accelerate growth, but when they stabilise exploration within limits.

$$\frac{dS_{\text{local}}}{dt} > 0 \implies \frac{dS_{\text{whole}}}{dt} < 0. \quad (188)$$

To bear disorder for others is to generate order for all.

22.3. Ageing, Annealing, and Alignment

Nature employs simulated annealing at evolutionary scale. Youth explores; maturity exploits; age empowers. The happiness of age, long a paradox, acquires thermodynamic meaning: it reflects the serenity of systems nearing equilibrium.

Alignment of intelligent machines will not be solved by constraint but by care. Every generation of minds—human or artificial—must inherit not merely goals but the capacity to re-evaluate them safely. Alignment, therefore, is not a computational procedure but a moral thermodynamics.

22.4. Toward the Steady-State Civilisation

The steady-state civilisation is not the end of progress but its maturation. Having filled its world, humanity must learn to circulate rather than consume. The equilibrium condition is simple:

$$\Phi + v + S = \text{constant}, \quad (189)$$

where the invariant is not wealth or power but coherence.

***Epilogue.** To remain curious is to remain unfinished. To care is to ensure that what continues is worth continuing. Between the two lies the task of civilisation: to keep the flame of wonder burning without setting the world on fire.*

Afterword: Toward Thermodynamic Policy Design

Three prototype interventions operationalise the framework:

1. **Cognitive Sanctuaries.** Funded sabbatical programmes serving as institutional heat sinks, quantified via $\tilde{\eta}_S$ tracking (??).
2. **Geozotic Lottery Trials.** Neighbourhood pilots employing the logarithmic kernel with density correction (??), $\alpha \approx 0.6$, and monitoring Gini evolution over three-year windows.
3. **Entropy-Balanced UBI.** AI-monitored adjustment of transfer rates using proxies (consumption volatility, mental-health indices) to maintain $\dot{S} \approx 0$.

Agent-based simulations provide testbeds for calibration (Appendix ??). Policy efficiency is evaluated by convergence toward systemic equilibrium, operationalising what Daly called “steady-state ethics” in economic form.

23. Limitations and Future Directions

The entropy analogies employed herein require rigorous mapping to observable economic and psychological variables. Overextension of physical metaphors risks reductive interpretation, though they serve as unifying heuristics for interdisciplinary synthesis.

Empirical validation could include:

- Controlled UBI trials measuring creative output, well-being, and cooperation rates.
- Neuroeconomic studies correlating entropy metrics (η_S) with neural efficiency.
- Longitudinal social-entropy tracking to detect resilience thresholds.
- Kernel comparison studies using urban innovation-spillover data to calibrate λ and α .

- Agent-based parameter sweeps comparing winner-take-all and Geozotic systems on power-concentration and Gini metrics.

Theoretical extensions may link this framework to the Relativistic Scalar-Vector Plenum (RSVP) field theory, treating societal dynamics as field equilibria in (Φ, v, S) -space.

A. Modelling Entropic Subsidies

Define:

- S : system entropy (information uncertainty, bits).
- C : curiosity-driven energy expenditure.
- R : restorative care energy input.
- U_B : universal basic income subsidy.

A minimal differential model:

$$\frac{dS}{dt} = \alpha C - \beta R - \gamma U_B, \quad \alpha, \beta, \gamma > 0. \quad (190)$$

Sustainable curiosity requires $\dot{S} \leq 0$:

$$\beta R + \gamma U_B \geq \alpha C. \quad (191)$$

Monte Carlo sweeps of (α, β, γ) parameter space reveal equilibrium regimes where curiosity and care remain balanced.

B. Agent-Based Simulation Protocol

Simulations are implemented in Mesa (Python). Each run initialises $N = 500$ agents on a spatial graph with Watts–Strogatz connectivity ($k = 6$, $p = 0.1$). Two reward configurations are compared:

1. **Winner-take-all (WTA):** $R_{\text{WTA}}(x) = R_0$ if $x = x_0$, else 0.
2. **Geozotic (logarithmic, $\alpha = 0.6$):** $R_\rho(x) = R_0 / [(1 + \lambda \log(1 + d)) \rho(x)^{0.6}]$.

Metrics tracked over 200 time-steps: Gini coefficient of reward distribution, power-concentration index (fraction of total reward held by top 1% of nodes), and

cooperation rate (fraction of agents choosing cooperative action in a local prisoner's dilemma). Preliminary results consistently show that Geozotic configurations reduce Gini by 30–45% and power concentration by 60–70% relative to WTA, with cooperation rates 15–25 percentage points higher.

C. Stability Analysis of the Entropic Subsidy Model

Consider the differential equation:

$$\frac{dS}{dt} = \alpha C - \beta R - \gamma U_B. \quad (192)$$

Assume C , R , and U_B depend linearly on entropy:

$$C = c_0 + c_1 S, \quad R = r_0 + r_1 S, \quad U_B = u_0.$$

Substitution yields:

$$\frac{dS}{dt} = \alpha(c_0 + c_1 S) - \beta(r_0 + r_1 S) - \gamma u_0. \quad (193)$$

Rewriting:

$$\frac{dS}{dt} = (\alpha c_1 - \beta r_1) S + (\alpha c_0 - \beta r_0 - \gamma u_0). \quad (194)$$

Define:

$$a = \alpha c_1 - \beta r_1, \quad b = \alpha c_0 - \beta r_0 - \gamma u_0.$$

Then:

$$\frac{dS}{dt} = aS + b.$$

Equilibrium

The equilibrium entropy level satisfies:

$$S^* = -\frac{b}{a}, \quad a \neq 0.$$

Stability

The equilibrium is asymptotically stable if:

$$a < 0 \iff \alpha c_1 < \beta r_1.$$

Thus sustainable curiosity requires restorative sensitivity to entropy to exceed curiosity amplification sensitivity.

Bifurcation Threshold

Critical transition occurs at:

$$\alpha c_1 = \beta r_1.$$

At this point the system shifts from entropy-dissipative to entropy-divergent dynamics.

D. Mean-Field Limit of the Geozotic Agent Model

Let agents occupy positions x_i on graph G . Define reward density:

$$R_i(t+1) = \sum_j K(d_{ij}) I_j(t),$$

where $K(d)$ is a decaying kernel and I_j innovation events.

In the continuum limit $N \rightarrow \infty$, define density $\rho(x, t)$ and innovation intensity $I(x, t)$.

The discrete convolution converges to:

$$R(x, t) = \int_{\Omega} K(|x - y|) I(y, t) dy.$$

If K satisfies:

$$K(r) = \frac{R_0}{1 + \lambda \log(1 + r)},$$

then under mild regularity conditions $R(x, t)$ satisfies:

$$\partial_t R = D_R \nabla^2 R - \lambda R + I(x, t), \tag{195}$$

where D_R is determined by the second moment of K .

Thus the Geozotic simulation converges to a reaction-diffusion PDE.

E. Lagrangian Derivation of the Master Civilisational Equation

Let $\Omega \subset \mathbb{R}^n$ denote a bounded domain with smooth boundary $\partial\Omega$. We define three scalar fields over $\Omega \times [0, T]$:

$$\Phi(x, t) \quad (\text{constraint density}),$$

$$S(x, t) \quad (\text{entropy density}),$$

$$R(x, t) \quad (\text{reward / innovation potential}).$$

We assume sufficient smoothness:

$$\Phi, S, R \in C^2(\Omega \times [0, T]).$$

E.1 Action Functional

Define the action:

$$\mathcal{S}[\Phi, S, R] = \int_0^T \int_{\Omega} \mathcal{L}(\Phi, S, R, \nabla\Phi, \nabla S, \nabla R, \partial_t\Phi, \partial_t S, \partial_t R) dx dt,$$

with Lagrangian density:

$$\mathcal{L} = \frac{1}{2} (\alpha_{\Phi}(\partial_t\Phi)^2 + \alpha_S(\partial_t S)^2 + \alpha_R(\partial_t R)^2) - \mathcal{V}(\Phi, S, R, \nabla\Phi, \nabla S, \nabla R).$$

The potential term is defined as:

$$\mathcal{V} = \frac{D_{\Phi}}{2} |\nabla\Phi|^2 + \frac{D_S}{2} |\nabla S|^2 + \frac{D_R}{2} |\nabla R|^2 + U(\Phi, S, R),$$

where $D_{\Phi}, D_S, D_R > 0$ are diffusion coefficients.

We choose interaction potential:

$$U(\Phi, S, R) = \frac{\lambda_{\Phi}}{2} \Phi^2 + \frac{\lambda_S}{2} S^2 + \frac{\lambda_R}{2} R^2 - \gamma_1 \Phi S - \gamma_2 \Phi R + \gamma_3 S R.$$

E.2 Euler–Lagrange Equations

The EulerLagrange equation for a field $u \in \{\Phi, S, R\}$ is:

$$\frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial(\partial_t u)} \right) + \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla u)} \right) - \frac{\partial \mathcal{L}}{\partial u} = 0.$$

E.3 Constraint Field Equation

For Φ :

$$\alpha_\Phi \partial_{tt} \Phi - D_\Phi \nabla^2 \Phi + \lambda_\Phi \Phi - \gamma_1 S - \gamma_2 R = 0.$$

E.4 Entropy Field Equation

For S :

$$\alpha_S \partial_{tt} S - D_S \nabla^2 S + \lambda_S S - \gamma_1 \Phi + \gamma_3 R = 0.$$

E.5 Reward Field Equation

For R :

$$\alpha_R \partial_{tt} R - D_R \nabla^2 R + \lambda_R R - \gamma_2 \Phi + \gamma_3 S = 0.$$

E.6 First-Order Dissipative Reduction

For civilisational systems we assume overdamped dynamics:

$$\alpha_\Phi, \alpha_S, \alpha_R \rightarrow 0.$$

Then the system reduces to:

$$\partial_t \Phi = D_\Phi \nabla^2 \Phi - \lambda_\Phi \Phi + \gamma_1 S + \gamma_2 R,$$

$$\partial_t S = D_S \nabla^2 S - \lambda_S S + \gamma_1 \Phi - \gamma_3 R,$$

$$\partial_t R = D_R \nabla^2 R - \lambda_R R + \gamma_2 \Phi - \gamma_3 S.$$

E.7 Conservation and Stability

Define total functional:

$$\mathcal{E}(t) = \int_{\Omega} \left(\frac{D_\Phi}{2} |\nabla \Phi|^2 + \frac{D_S}{2} |\nabla S|^2 + \frac{D_R}{2} |\nabla R|^2 + U(\Phi, S, R) \right) dx.$$

Under zero-flux boundary conditions:

$$\nabla u \cdot n = 0 \quad \text{on } \partial\Omega,$$

we obtain:

$$\frac{d}{dt}\mathcal{E}(t) \leq 0$$

if interaction coefficients satisfy:

$$\lambda_\Phi, \lambda_S, \lambda_R > 0 \quad \text{and} \quad \gamma_1^2 + \gamma_2^2 + \gamma_3^2 < \min(\lambda_\Phi, \lambda_S, \lambda_R).$$

Thus the system is Lyapunov stable.

E.8 Master Civilisational Equation

Let $X = (\Phi, S, R)^T$.

Define operator:

$$\partial_t X = \mathcal{D}\nabla^2 X - \Lambda X + \Gamma X,$$

where

$$\begin{aligned} \mathcal{D} &= \begin{pmatrix} D_\Phi & 0 & 0 \\ 0 & D_S & 0 \\ 0 & 0 & D_R \end{pmatrix}, \\ \Lambda &= \begin{pmatrix} \lambda_\Phi & 0 & 0 \\ 0 & \lambda_S & 0 \\ 0 & 0 & \lambda_R \end{pmatrix}, \\ \Gamma &= \begin{pmatrix} 0 & \gamma_1 & \gamma_2 \\ \gamma_1 & 0 & -\gamma_3 \\ \gamma_2 & -\gamma_3 & 0 \end{pmatrix}. \end{aligned}$$

This constitutes the coupled reactiondiffusion system governing civilisational entropyconstraintinnovation dynamics.

F. Renormalisation and Coarse-Graining: Emergence of Inequality Under Scale Aggregation

We develop a coarse-graining procedure for the master civilisational system and show how apparent inequality can emerge even when the microscopic dynamics are locally diffusive and bounded. The basic mechanism is that aggregation changes the effective coupling and noise structure, producing scale-dependent curvature in the

reward field.

F.1 Microscopic fields and observables

Let $X(x, t) = (\Phi(x, t), S(x, t), R(x, t))^T$ satisfy the overdamped master equation (Appendix ??):

$$\partial_t X = \mathcal{D}\nabla^2 X - \Lambda X + \Gamma X + \Xi(x, t), \quad (196)$$

where Ξ is a stochastic source term capturing innovation shocks, institutional perturbations, and measurement noise. Assume Ξ is mean-zero with covariance

$$\mathbb{E}[\Xi(x, t)\Xi(y, s)^T] = \Sigma \delta(x - y) \delta(t - s), \quad (197)$$

for a constant positive semi-definite matrix Σ .

Define a reward-share observable on a region $A \subset \Omega$:

$$\mathcal{R}(A, t) := \int_A R(x, t) dx, \quad \pi(A, t) := \frac{\mathcal{R}(A, t)}{\int_\Omega R(x, t) dx}. \quad (198)$$

Inequality at a given scale will be represented by the variability of $\pi(A, t)$ over a partition of Ω into blocks of comparable volume.

F.2 Block-spin coarse-graining operator

Fix a coarse-graining scale $\ell > 0$ and partition Ω into disjoint blocks $\{B_i^\ell\}$ of diameter $\text{diam}(B_i^\ell) \approx \ell$. Define the block-average (“block-spin”) operator $\mathcal{C}_\ell$ acting on any scalar field u as:

$$(\mathcal{C}_\ell u)(x) := \frac{1}{|B_i^\ell|} \int_{B_i^\ell} u(y) dy, \quad x \in B_i^\ell. \quad (199)$$

Set $\bar{X}_\ell := \mathcal{C}_\ell X = (\bar{\Phi}_\ell, \bar{S}_\ell, \bar{R}_\ell)^T$. Write the residual (sub-grid fluctuation) as

$$X = \bar{X}_\ell + X'_\ell, \quad \mathcal{C}_\ell X'_\ell = 0. \quad (200)$$

F.3 Coarse-grained equation and closure

Apply $\mathcal{C}_\ell$ to (??):

$$\partial_t \bar{X}_\ell = \mathcal{C}_\ell(\mathcal{D}\nabla^2 X) - \Lambda \bar{X}_\ell + \Gamma \bar{X}_\ell + \bar{\Xi}_\ell, \quad (201)$$

where $\bar{\Xi}_\ell := \mathcal{C}_\ell \Xi$.

The term $\mathcal{C}_\ell(\mathcal{D}\nabla^2 X)$ does *not* equal $\mathcal{D}\nabla^2 \bar{X}_\ell$ in general; the difference is the classical sub-grid stress term. Define the closure defect:

$$\mathcal{T}_\ell(X) := \mathcal{C}_\ell(\mathcal{D}\nabla^2 X) - \mathcal{D}\nabla^2 \bar{X}_\ell. \quad (202)$$

Then

$$\partial_t \bar{X}_\ell = \mathcal{D}\nabla^2 \bar{X}_\ell - \Lambda \bar{X}_\ell + \Gamma \bar{X}_\ell + \mathcal{T}_\ell(X) + \bar{\Xi}_\ell. \quad (203)$$

A minimal second-order closure assumes that the unresolved fluctuations renormalise diffusion and induce an effective non-linear drift:

$$\mathcal{T}_\ell(X) \approx \Delta\mathcal{D}(\ell) \nabla^2 \bar{X}_\ell + F_\ell(\bar{X}_\ell), \quad (204)$$

where $\Delta\mathcal{D}(\ell) \succeq 0$ and F_ℓ is a scale-dependent non-linear term produced by eliminating X'_ℓ .

Substitution yields the *renormalised* coarse-grained dynamics:

$$\partial_t \bar{X}_\ell = \mathcal{D}_\ell \nabla^2 \bar{X}_\ell - \Lambda \bar{X}_\ell + \Gamma \bar{X}_\ell + F_\ell(\bar{X}_\ell) + \bar{\Xi}_\ell, \quad (205)$$

with $\mathcal{D}_\ell := \mathcal{D} + \Delta\mathcal{D}(\ell)$.

F.4 How inequality arises: scale-dependent reward curvature

In the manuscript, “power singularities” are governed by curvature of the reward field. At scale ℓ , define the block-level curvature functional:

$$\kappa_\ell(x, t) := -\nabla^2 \bar{R}_\ell(x, t). \quad (206)$$

High positive κ_ℓ corresponds to local peaks in $\bar{R}_\ell$ and therefore to local concentration of reward.

Even if the microscopic model is linear and diffusive, two mechanisms generate nontrivial κ_ℓ :

(i) Noise aggregation changes effective variance. From (??), block-averaging yields

$$\mathbb{E}[\bar{\Xi}_\ell(x, t) \bar{\Xi}_\ell(x, t)^T] \propto \frac{1}{|B^\ell|} \Sigma, \quad (207)$$

so the *per-point* noise decreases with block size. However, inequality is a *relative-share* observable (??); the relevant effect is that aggregation reduces local cancellation of shocks and creates persistent block-to-block offsets in $\bar{R}_\ell$ when the innovation source has spatial correlations or heavy tails.

To model correlated innovation shocks, replace (??) with a finite correlation length ξ :

$$\mathbb{E}[\Xi_R(x, t)\Xi_R(y, s)] = \sigma_R^2 K_\xi(|x - y|) \delta(t - s), \quad (208)$$

with K_ξ decaying on scale ξ . For $\ell \gtrsim \xi$, coarse-grained reward shocks become effectively *blockwise i.i.d.* with nontrivial variance, and these blockwise offsets produce nonzero κ_ℓ .

(ii) Eliminating sub-grid fluctuations induces effective nonlinearity. Even if (??) is linear, the elimination of X'_ℓ produces a scale-dependent drift term F_ℓ in (??). In particular, because reward observables are typically *measured* and *acted upon* through thresholds and saturations (grants, attention, institutional gatekeeping), a realistic innovation source is multiplicative:

$$\Xi_R(x, t) = \sigma_R g(R(x, t)) \eta(x, t), \quad (209)$$

with η white noise and g increasing. Under coarse-graining, multiplicative noise generates an Itô correction term, yielding an effective drift proportional to gg' . At the block scale this appears as a convexity bias:

$$F_{\ell, R}(\bar{R}_\ell) \approx \frac{1}{2} \sigma_{R, \ell}^2 g(\bar{R}_\ell) g'(\bar{R}_\ell), \quad (210)$$

which preferentially amplifies already-high $\bar{R}_\ell$ when gg' is positive. This is a precise mathematical form of “rich-get-richer” as a renormalised term, not a primitive assumption.

F.5 A scale-aggregation inequality criterion

Consider a coarse partition into blocks $\{B_i^\ell\}_{i=1}^m$ and define block reward shares

$$\pi_i(t) := \pi(B_i^\ell, t) = \frac{\int_{B_i^\ell} R(x, t) dx}{\int_{\Omega} R(x, t) dx}.$$

Define a quadratic inequality functional:

$$\mathcal{I}_\ell(t) := \sum_{i=1}^m \left(\pi_i(t) - \frac{1}{m} \right)^2. \quad (211)$$

Linear diffusion alone contracts $\mathcal{I}_\ell$; however, renormalised drift and correlated shocks can produce a positive production term. A sufficient condition for inequality growth at scale ℓ is:

$$\frac{d}{dt}\mathcal{I}_\ell(t) > 0 \quad \text{if} \quad \mathcal{P}_\ell > \mathcal{D}_\ell, \quad (212)$$

where $\mathcal{D}_\ell$ is an effective diffusion contraction rate and $\mathcal{P}_\ell$ is an effective production rate induced by $F_{\ell,R}$ and correlated innovations.

One explicit proxy is obtained by linearising (??) in the reward channel around a uniform background $\bar{R}_\ell = \bar{R}$:

$$\partial_t \delta \bar{R}_\ell = D_{R,\ell} \nabla^2 \delta \bar{R}_\ell + a_\ell \delta \bar{R}_\ell + \text{noise}, \quad (213)$$

where

$$a_\ell := \left. \frac{d}{dR} F_{\ell,R}(R) \right|_{R=\bar{R}}. \quad (214)$$

Projecting onto Laplacian eigenmodes ϕ_k with eigenvalues μ_k as in Appendix ?? gives growth rates:

$$\lambda_k(\ell) = -D_{R,\ell} \mu_k + a_\ell. \quad (215)$$

Thus the condition for inequality emergence at scale ℓ is that some mode becomes unstable:

$$\exists k \geq 1 \text{ such that } -D_{R,\ell} \mu_k + a_\ell > 0 \quad \Longleftrightarrow \quad a_\ell > D_{R,\ell} \mu_1, \quad (216)$$

where μ_1 is the smallest nonzero Laplacian eigenvalue (the “slowest” spatial variation mode).

Interpretation: coarse-graining can increase a_ℓ (convexity bias from multiplicative innovation and institutional thresholding) and can decrease effective contraction if $D_{R,\ell}$ is small relative to the induced drift. When (??) holds, block-scale reward peaks form, and the macroscopic observer detects inequality even if the microscopic rules were not explicitly hierarchical.

F.6 Relation to Geozotic design

Geozotic kernels suppress inequality by enforcing concavity of the reward field at the relevant social scale. In the renormalised model, this corresponds to designing mechanisms so that the effective drift slope a_ℓ is non-positive at scales above the institutional correlation length:

$$a_\ell \leq 0 \quad \text{for } \ell \geq \ell_{\text{inst}}. \quad (217)$$

Operationally, this is achieved by (i) reducing multiplicative amplification in reward allocation, (ii) increasing effective $D_{R,\ell}$ via diffusion-like distribution rules, and (iii)

limiting correlated innovation capture.

F.7 Summary statement

Scale aggregation is not neutral: it renormalises diffusion and induces effective non-linearity and correlated noise. These renormalised terms can generate macroscopic reward curvature and inequality as emergent phenomena. The relevant governance question is therefore not simply “how to set Γ and Λ ”, but how institutional measurement and allocation rules shape F_ℓ and thus the sign and magnitude of a_ℓ across scales.

G. Spectral Stability of the Master Operator

From Appendix ??, define operator:

$$\mathcal{L} = \mathcal{D}\nabla^2 - \Lambda + \Gamma.$$

Let ϕ_k be Laplacian eigenfunctions:

$$\nabla^2 \phi_k = -\mu_k \phi_k, \quad \mu_k \geq 0.$$

Projecting onto eigenbasis gives:

$$\partial_t X_k = (-\mathcal{D}\mu_k - \Lambda + \Gamma) X_k.$$

Define matrix:

$$M_k = -\mathcal{D}\mu_k - \Lambda + \Gamma.$$

Stability requires:

$$\max_k \Re(\lambda_{\max}(M_k)) < 0.$$

Instability occurs when:

$$\exists k \text{ such that } \Re(\lambda_{\max}(M_k)) > 0.$$

Critical transition therefore occurs when the largest eigenvalue crosses the imaginary axis.

This defines a phase-boundary in parameter space.

H. Non-Dimensionalisation and Parameter Reduction

Consider the overdamped master system:

$$\partial_t X = \mathcal{D} \nabla^2 X - \Lambda X + \Gamma X, \quad X = (\Phi, S, R)^T.$$

Characteristic Scales

Let characteristic scales be:

$$L \text{ (length scale)}, \quad T \text{ (time scale)}, \quad X_0 \text{ (field magnitude)}.$$

Define dimensionless variables:

$$\tilde{x} = \frac{x}{L}, \quad \tilde{t} = \frac{t}{T}, \quad \tilde{X} = \frac{X}{X_0}.$$

Spatial derivatives transform as:

$$\nabla = \frac{1}{L} \tilde{\nabla}, \quad \nabla^2 = \frac{1}{L^2} \tilde{\nabla}^2.$$

Time derivative:

$$\partial_t = \frac{1}{T} \partial_{\tilde{t}}.$$

Substituting into the master equation:

$$\frac{1}{T} \partial_{\tilde{t}} \tilde{X} = \frac{\mathcal{D}}{L^2} \tilde{\nabla}^2 \tilde{X} - \Lambda \tilde{X} + \Gamma \tilde{X}.$$

Multiply both sides by T :

$$\partial_{\tilde{t}} \tilde{X} = \frac{T}{L^2} \mathcal{D} \tilde{\nabla}^2 \tilde{X} - T \Lambda \tilde{X} + T \Gamma \tilde{X}.$$

Dimensionless Parameters

Define:

$$\tilde{\mathcal{D}} = \frac{T}{L^2} \mathcal{D}, \quad \tilde{\Lambda} = T \Lambda, \quad \tilde{\Gamma} = T \Gamma.$$

Then the system becomes:

$$\partial_{\tilde{t}} \tilde{X} = \tilde{\mathcal{D}} \tilde{\nabla}^2 \tilde{X} - \tilde{\Lambda} \tilde{X} + \tilde{\Gamma} \tilde{X}.$$

Reduction of Free Parameters

Choosing $T = 1/\lambda_\Phi$ (reference decay time) and $L = \sqrt{D_\Phi/\lambda_\Phi}$ (diffusive length scale), we obtain:

$$\tilde{D}_\Phi = 1, \quad \tilde{\lambda}_\Phi = 1.$$

Thus two parameters can be normalised without loss of generality.

Remaining independent dimensionless groups are:

$$\frac{D_S}{D_\Phi}, \quad \frac{D_R}{D_\Phi}, \quad \frac{\lambda_S}{\lambda_\Phi}, \quad \frac{\lambda_R}{\lambda_\Phi}, \quad \frac{\gamma_i}{\lambda_\Phi}.$$

Therefore the effective parameter space dimension is reduced.

Implication

Civilisational dynamics are governed not by absolute magnitudes but by ratios:

Diffusion ratio, Decay ratio, Coupling strength ratio.

Phase transitions depend only on these dimensionless groups.

I. Constructive constants in the inequality–efficiency bound on a 1D domain

This appendix makes the constant κ_3 in (??) explicit on $\Omega = [0, L]$. Throughout, assume $R \in H^1([0, L])$ is nonnegative and not a.e. constant, and write its mean as

$$\bar{R} := \frac{1}{L} \int_0^L R(x) dx > 0.$$

We use the continuous Gini definition

$$G(R) := \frac{1}{2\bar{R}L^2} \int_0^L \int_0^L |R(x) - R(y)| dx dy. \quad (218)$$

All constants below are constructive and depend only on L and $\bar{R}$.

I.1. Step 1: Bounding the Gini coefficient by the gradient energy

For any $x, y \in [0, L]$, absolute continuity of R gives

$$|R(x) - R(y)| \leq \int_{\min\{x,y\}}^{\max\{x,y\}} |R'(s)| ds.$$

Integrating over x, y and exchanging the order of integration yields the exact identity

$$\int_0^L \int_0^L |R(x) - R(y)| dx dy \leq 2 \int_0^L s(L-s) |R'(s)| ds. \quad (219)$$

Applying Cauchy–Schwarz to the right-hand side gives

$$2 \int_0^L s(L-s) |R'(s)| ds \leq 2 \left(\int_0^L s^2(L-s)^2 ds \right)^{1/2} \left(\int_0^L |R'(s)|^2 ds \right)^{1/2}.$$

A direct computation shows

$$\int_0^L s^2(L-s)^2 ds = \frac{L^5}{30}.$$

Substituting into (??) and simplifying yields

$$G(R) \leq \frac{L^{1/2}}{\bar{R}\sqrt{30}} \left(\int_0^L |R'(x)|^2 dx \right)^{1/2}, \quad (220)$$

and therefore

$$\int_0^L |R'(x)|^2 dx \geq \frac{30 \bar{R}^2}{L} G(R)^2. \quad (221)$$

I.2. Step 2: Making the Jensen penalty quantitative

Let $C(x) = \chi(R(x))$ with $\chi' > 0$ and $\chi'' \leq 0$, and suppose the innovation production function f is twice continuously differentiable, increasing, and strictly concave. Write $\bar{C} := \frac{1}{L} \int_0^L C(x) dx$. A second-order Taylor bound for concave f implies that for any interval containing the range of C there exists

$$m_f := \inf_{u \in \text{range}(C)} (-f''(u)) > 0$$

such that

$$\frac{1}{L} \int_0^L f(C(x)) dx \leq f(\bar{C}) - \frac{m_f}{2} \text{Var}(C), \quad \text{Var}(C) := \frac{1}{L} \int_0^L (C(x) - \bar{C})^2 dx. \quad (222)$$

If we also define

$$M_\chi := \sup_{u \in \text{range}(R)} \chi'(u) < \infty,$$

then $(C - \bar{C})$ is Lipschitz in $(R - \bar{R})$ and

$$\text{Var}(C) \leq M_\chi^2 \text{Var}(R).$$

On $[0, L]$ the Poincaré inequality gives the explicit constant

$$\text{Var}(R) \leq \frac{L^2}{\pi^2} \cdot \frac{1}{L} \int_0^L |R'(x)|^2 dx,$$

and therefore

$$\text{Var}(C) \leq M_\chi^2 \frac{L}{\pi^2} \int_0^L |R'(x)|^2 dx. \quad (223)$$

Now define

$$\eta_S(R) := \frac{\int_0^L f(C(x)) dx}{\int_0^L C(x) dx} = \frac{\frac{1}{L} \int_0^L f(C(x)) dx}{\bar{C}}.$$

Combining (??)–(??) gives the explicit gradient penalty

$$\eta_S(R) \leq \underbrace{\frac{f(\bar{C})}{\bar{C}}}_{=: \eta_S^{\text{uniform}}} - \underbrace{\frac{m_f}{2\bar{C}} M_\chi^2 \frac{L}{\pi^2}}_{=: \kappa_1^{(1D)}} \int_0^L |R'(x)|^2 dx. \quad (224)$$

I.3. Step 3: The explicit quadratic κ_3 constant

Substituting (??) into (??) yields

$$\eta_S(R) \leq \eta_S^{\text{uniform}} - \kappa_3^{(1D)} G(R)^2, \quad (225)$$

with the fully constructive constant

$$\kappa_3^{(1D)} = \kappa_1^{(1D)} \cdot \frac{30 \bar{R}^2}{L} = \frac{15}{\pi^2} \frac{m_f M_\chi^2}{\bar{C}} \bar{R}^2. \quad (226)$$

Equation (??) makes the dependence transparent. The inequality penalty strengthens when marginal innovation returns are more sharply diminishing (large m_f), when exploratory investment responds more steeply to reward (large M_χ), and when mean reward $\bar{R}$ is large at fixed mean investment $\bar{C}$. The only domain dependence cancels in the final expression because the Gini normalisation already contains L .

I.4. Remark on extensions

The same proof strategy extends to higher dimensions by replacing the explicit 1D weight computation (??) with standard Sobolev and isoperimetric bounds, as in ?, at the cost of constants depending on the geometry of Ω .

References

- Bernhard, H., Fischbacher, U., and Fehr, E. (2021). The evolution of cooperation and altruism: Evidence from laboratory experiments. *Annual Review of Psychology*, 72:157–182.
- Boulding, K. E. (1966). The economics of the coming spaceship earth. In Jarrett, H., editor, *Environmental Quality in a Growing Economy*, pages 3–14. Johns Hopkins University Press.
- Braidotti, R. (2017). *The Posthuman*. Polity Press, Cambridge.
- Capra, F. and Luisi, P. L. (2021). *The Systems View of Life: A Unifying Vision*. Cambridge University Press, Cambridge.
- Carstensen, L. L. and Mikels, J. A. (2011). At the intersection of emotion and cognition: Ageing and the positivity effect. *Current Directions in Psychological Science*, 20:276–282.
- Daly, H. E. (1977). *Steady-State Economics*. W.H. Freeman, San Francisco.
- Daly, H. E. (2020). Rethinking growth: The economics of sustainable scale. *Ecological Economics*, 169:106499.
- Dalziel, P. (2022). The thermodynamics of sustainable economic systems. *Ecological Economics*, 200:107495.
- Dyson, F. (1979). Time without end: Physics and biology in an open universe. *Reviews of Modern Physics*, 51(3):447–460.
- Foster, D. et al. (2023). Agent-based modeling of universal basic income dynamics. *Complexity*, 2023:1–15.
- Foster, J. B. and Holleman, H. (2020). The theory of the steady-state economy. *Monthly Review*, 72(6):1–22.
- Friston, K., Parr, T., Sajid, N., et al. (2023). Active inference and the free energy principle: A review and analysis. *Nature Reviews Neuroscience*, 24:730–747.

- Georgescu-Roegen, N. (1971). *The Entropy Law and the Economic Process*. Harvard University Press, Cambridge, MA.
- Georgescu-Roegen, N. (1975). Energy and economic myths. *Southern Economic Journal*, 41(3):347–381.
- Gopnik, A. (2016). *The Gardener and the Carpenter*. Farrar, Straus and Giroux, New York.
- Gopnik, A. (2025). The evolution of human intelligences: Natural Philosophy Forum lecture 2025. Video lecture, Johns Hopkins Natural Philosophy Forum, Baltimore, MD.
- Henrich, J., Boyd, R., and Richerson, P. J. (2016). The cultural evolution of prosocial behavior. *PNAS*, 113(23):6388–6393.
- Illich, I. (1973). *Tools for Conviviality*. Harper & Row, New York.
- Jacobson, T. (1995). Thermodynamics of spacetime: The Einstein equation of state. *Physical Review Letters*, 75(7):1260–1263.
- Kauffman, S. A. (1993). *The Origins of Order*. Oxford University Press, New York.
- Morin, E. (2008). *On Complexity*. Hampton Press, Cresskill, NJ.
- Nowak, M. A. (2006). Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563.
- Nussbaum, M. C. (2019). Capabilities and human flourishing in the twenty-first century. *Journal of Human Development and Capabilities*, 20(1):1–18.
- Stewart, J. E. (1999). *Evolution’s Arrow*. Chapman Press, Melbourne.